

STRONG FINITE-TYPE CONDITIONS UNDER POLYNOMIAL AND FORMAL POWER-SERIES EXTENSION

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ABSTRACT. This document is the working log accompanying the main paper of Tran–Toan–Vo–Nguyen, which constructs a one-dimensional local SFT domain R for which $R[X]$ is not SFT [22]. It preserves the extended proofs, variants, structural reductions, formal power-series investigations, dimension calculations, and prime-chain constructions developed around that example, with published input cited and project results identified explicitly.

1. MAP OF THE PROVED RESULTS

All rings are commutative with identity. The symbol X denotes a polynomial variable and Y a formal variable. A formal heading or summary entry marked [Project result] is proved within this project rather than quoted as a literature theorem. This marker records provenance, not a claim of historical priority beyond the sources cited here. Results imported from the literature carry a citation next to the statement; citations attached to a project result identify published inputs used in its proof.

1.1. Notable advances on earlier project questions. Here “earlier” refers to questions left open in earlier stages of this project, not to a claim that they were previously open throughout the literature. The strongest new conclusions and their remaining boundaries are summarized below; unresolved global questions are collected in Table 6.

Table 1: Notable project advances relative to earlier open entries.

Earlier project question	Current status
What are the SFT, t -SFT, and VSFT statuses of $R_p[[Y]]$ and $R_0[[Y]]$?	Closed in this project. Theorem 4.14 constructs, inside each formal ring, an ideal that is not t -SFT. Consequently neither ring is t -SFT, SFT, or VSFT. This settles the entries that were previously unknown without using an invalid substitution of a formal series at $Y = u$; see Remark 4.15. [Project result]
Are the ordinary Krull dimensions of these two formal rings finite?	Closed, with a cardinal strengthening. Theorem 6.12 constructs an actual prime chain of cardinality at least $2^{\aleph_1}$ in each ring. Thus both have infinite ordinary Krull dimension. Under CH their cardinal chain dimensions equal $2^{\aleph_1}$; the exact value in ZFC remains open. [Project result]
Does t -SFT ascend from a domain D to $D[X]$?	Closed negatively in general. The coefficient rings R_p and R_0 are t -SFT, whereas their polynomial rings are not t -SFT by Theorem 4.9. The examples are nonnormal, so the integrally closed positive theorem theorem 4.3 remains a genuine boundary. [Project result]

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Earlier project question	Current status
How can one recognize rings for which $R[X]$ is SFT?	Reduced exactly, but not intrinsically classified. Theorem 2.15 gives an equivalence in terms of uppers to zero over every residue domain. The residual N-1 theorem and its consequences Theorem 3.14 and Corollary 3.15 further show that a finite-dimensional normal counterexample must have a nonzero residue domain with nonfinite normalization. [Project result]
Does $R[[X]]$ SFT imply $R[[X, Y]]$ SFT?	Still open, but substantially narrowed. It is enough to solve the problem for domains by Corollary 5.2. Moreover, every bad residue-domain reduction must have dimension at least two by Theorem 5.23; in particular the implication holds when $\dim R[[X]] \leq 1$ by Corollary 5.24. [Project result]
Can a finite-dimensional normal SFT domain have an infinite-dimensional one-variable formal ring?	Still open, with an exact fibre criterion. Theorem 5.13 proves that finite formal dimension is equivalent to one uniform finite bound on all residue generic formal fibres. The valuation-conductor recursion Theorem 5.3 and Corollary 5.8 sharply restrict any possible normal counterexample. [Project result]

The first two rows close entries that were explicitly unresolved in an earlier draft of this compendium. They should not be read as a claim that this project discovered the general failure of zero-to-one formal SFT ascent, which was already known [7]. The contribution here is the direct simultaneous status determination for these explicit one-dimensional coefficient rings and the huge-chain strengthening. The published Toan–Kang theorem [20, 21] assumes a non-SFT coefficient ring, so it does not yield the direct one-variable chains for R_p and R_0 , which are SFT. The direct chain theorem is a project result that has not received external peer review.

1.2. Constructed rings. Tran–Toan–Vo–Nguyen construct, for each prime p , a one-dimensional local domain R_p of characteristic p and a one-dimensional local $\mathbf{Q}$ -algebra R_0 , both having a non-SFT upper to zero in the polynomial ring [22, Theorems 4.1 and 5.1]. The table records the complete status proved around those examples in this project, including the t -SFT and VSFT distinctions, exact polynomial dimensions, formal-series obstructions, and cardinal prime chains.

Table 2: Exact status of the explicit examples.

Ring	Dimension	SFT / t -SFT / VSFT	Further structure
R_p	1	yes / yes / no	local, nonnormal, non-Noetherian [Project result]
$R_p[X]$	3	no / no / no	bad non-finitely-generated upper to zero [Project result]
$R_p[[Y]]$	∞	no / no / no	actual prime chain of size $\geq 2^{\aleph_1}$ [Project result]
R_0	1	yes / yes / yes	local $\mathbf{Q}$ -algebra, nonnormal, non-Noetherian [Project result]
$R_0[X]$	3	no / no / no	bad non-finitely-generated upper to zero [Project result]
$R_0[[Y]]$	∞	no / no / no	actual prime chain of size $\geq 2^{\aleph_1}$ [Project result]

The formal entries are direct conclusions, not transfers of polynomial evaluation. The ideal in Theorem 4.14 proves that $R_p[[Y]]$ and $R_0[[Y]]$ are not t -SFT; hence they are neither SFT nor VSFT.

Two further consequences should not be confused with that direct proof. If S is one of the coefficient rings for which $S[X]$ is non-SFT, then

$$S[[Y_1, \dots, Y_s]][X] \quad (s \geq 1)$$

is non-SFT by quotienting by $(Y_1, \dots, Y_s)$. On the other hand, with $A = S[X]$, the Toan–Kang theorem [20, 21] applied to the non-SFT coefficient ring A gives

$$A[[Y]] \text{ is non-SFT,} \quad A[[Y]] \text{ has a prime chain of size at least } 2^{\aleph_1}.$$

This hybrid conclusion is independent of the stronger direct theorem for $S[[Y]]$.

1.3. Polynomial-extension results.

Table 3: Polynomial results proved or used in the project.

Hypothesis	Conclusion
R is any ring	$R[X]$ is SFT exactly when R is SFT and every upper to zero over every residue domain $(R/P)[X]$ is SFT. [Project result]
D is a one-dimensional SFT domain	Polynomial stability is equivalent to the SFT property for every upper to zero of $D[X]$. [Project result]
D is one-dimensional, normal, and SFT	$D[X]$ is SFT [18].
D is normal, SFT, and a going-down domain	$D[X]$ is SFT [18][4, Theorem 4.1].
D is finite-dimensional, SFT, and residually N-1	$D[X]$ is SFT. [Project result]
D is finite-dimensional, normal, and every nonzero residue domain has finite normalization	$D[X]$ is SFT; in dimension two it is enough to test height-one residue domains. [Project result]
$D[X]$ is t -SFT	D is t -SFT, by exact contraction of the t - and v -closures ([Project result]); if D is integrally closed, the converse is Kang–Park [14, Theorem 2.4].
$U = k[Z_1, Z_2, \dots]$	U is a UFD and is t -SFT with exponent-one witnesses, but the ideal $(Z_1, Z_2, \dots)$ is not SFT. [Project result]
$R = R_p$ or R_0	$R[X_1, \dots, X_m]$ is non-SFT for every $m \geq 1$. [Project result]
$R = R_p$ or R_0	$\dim R[X] = 3$. [Project result]

1.4. Power-series and dimension results. The formal-series theory is organized by horizontal primes and generic formal fibres. For a finite-dimensional SFT ring A , write

$$G_P^{(s)} = ((A/P)[[X_1, \dots, X_s]])_{(A/P) \setminus \{0\}}, \quad g_P^{(s)} = \dim G_P^{(s)}, \quad g_s = \sup_P g_P^{(s)}.$$

The main proved conclusions are summarized below.

Table 4: Formal-series and Krull-dimension results.

Setting	Proved conclusion
A is SFT	$A[[Y]]$ is SFT exactly when every zero-contraction prime over every residue domain is SFT. [Project result]
$D[[Y]]$ is t -SFT	D is t -SFT; the proof uses $(FD[[Y]])_v \cap \text{Frac}(D) = F_v$ and $(ID[[Y]])_t \cap \text{Frac}(D) = I_t$, not quotient permanence. [Project result]
Universal $R[[X]] \Rightarrow R[[X, Y]]$ question	It reduces exactly to the case of coefficient domains. [Project result]
A is finite-dimensional and SFT	$g_s \leq \dim A[[\mathbf{X}]] \leq d + (d + 1)g_s$; in particular the formal ring is finite-dimensional exactly when the residue generic-fibre dimensions admit one uniform finite bound. [Project result]
$A \subseteq B$ is module-finite	SFT is invariant; the same is true after adjoining any fixed finite collection of polynomial and formal variables, and the corresponding formal-series rings have equal dimension. [Project result]
$A \subseteq B$ is uniformly radical of bounded exponent in characteristic p	SFT ascends to every finite mixed polynomial/formal extension. [Project result]
R is normal, finite-dimensional, SFT, and has a proper valuation overring with nonzero conductor P	$g_s(R) = \max\{s, g_s(R/P)\}$; finite formal dimension reduces to the smaller normal quotient. [Project result]
D is a residue domain with $\dim D \leq 1$ arising in the one-step problem	$D[[Y]]$ is SFT; hence a bad $1 \Rightarrow 2$ prime must occur over a residue domain of dimension at least two. [Project result]
$R = R_p$ or R_0	$R[[Y]]$ is neither t -SFT, SFT, nor VSFT; it has an actual prime chain of cardinality at least $2^{\aleph_1}$ and therefore infinite ordinary dimension. [Project result]
D is non-SFT	The published Toan–Kang theorem gives an actual prime chain of cardinality at least $2^{\aleph_1}$ in $D[[Y]]$ [20, 21].
Zero-dimensional SFT coefficients, at least two variables	Going up, incomparability, and integrality reduce exactly to the corresponding residue-field bounded-radical/finite conditions. [Project result]

1.5. **Cardinality snapshot.** The notation cdim denotes the supremum of cardinalities of actual chains in the spectrum; it is not ordinary numerical Krull dimension.

Table 5: Cardinality results proved or quoted in the project.

Ring or hypothesis	Cardinal conclusion
R_p, R_0	Both rings are countable. [Project result]
$R_p[[\mathbf{Y}]], R_0[[\mathbf{Y}]]$, with a nonempty finite tuple $\mathbf{Y}$	The ring cardinality is $\mathfrak{c}$ and $\text{cdim} \geq 2^{\aleph_1}$; under CH both cdim and the spectrum cardinality equal $2^{\mathfrak{c}} = 2^{\aleph_1}$. [Project result]
D non-SFT	$D[[Y]]$ has an actual chain of at least $2^{\aleph_1}$ primes [20, 21].
Countable valuation example V	V is a one-dimensional non-SFT domain; $ V[[Y]] = \mathfrak{c}$ and $\text{cdim } V[[Y]] \geq 2^{\aleph_1}$ by Toan–Kang [20, 21]. [Project result]

Ring or hypothesis	Cardinal conclusion
Nagata realization	Starting from Nagata's published example [19, Tag 02JC], there is a countable Noetherian (hence VSFT) ring R ; $R[[Y]]$ is again Noetherian and VSFT, and $\text{cdim } R[[Y]] = \aleph_0$ is an unattained supremum. [Project result]
CH valuation realization	There is an infinite-dimensional VSFT valuation domain R for which $R[[Y]]$ is SFT and $\text{cdim } R[[Y]] = \aleph_1$ [15]. [Project result]

2. DEFINITIONS AND PERMANENCE MECHANISMS

2.1. SFT data and the prime criterion.

Definition 2.1. The SFT terminology is Arnold's [3]; the SFT-data and VSFT formulations follow Coykendall–Dutta [8, Definitions 1.2–1.3 and 2.3–2.4]. Let I be an ideal of a ring A .

- (i) The ideal I is *strong finite type*, or *SFT*, if there are a finitely generated ideal $B \subseteq I$ and an integer $n \geq 1$ such that

$$x^n \in B \quad (x \in I).$$

We call (I, B, n) SFT data.

- (ii) The ideal I is *very strong finite type*, or *VSFT*, if there are such B, n with the stronger containment $I^n \subseteq B$.
- (iii) A ring is SFT, respectively VSFT, if every ideal has the indicated property.

The distinction in Definition 2.1 is essential. Elementwise power containment need not imply containment of an ideal power. Arnold's Cohen-type criterion will be used throughout.

Theorem 2.2 (Arnold's prime criterion [3, Proposition 2.2]). *A ring is SFT if and only if each of its prime ideals is an SFT ideal.*

Proof. The nontrivial direction is the cited prime-ideal criterion. The reverse direction is immediate from the definition of an SFT ring. We record the result explicitly because several later proofs construct data only for primes. $\square$

Lemma 2.3 (Standard consequences [3, Proposition 2.5] [8, Proposition 2.9]). *Let (I, B, n) be SFT data in A .*

- (a) (I, B, N) is SFT data for every $N \geq n$.
- (b) $\sqrt{I} = \sqrt{B}$.
- (c) Every finitely generated ideal is SFT; hence every Noetherian ring is SFT.
- (d) $\text{Spec } A$ is a Noetherian topological space whenever A is SFT.

Proof. If $x \in I$ and $N \geq n$, then $x^N = x^{N-n}x^n \in B$, proving (a). The inclusion $\sqrt{B} \subseteq \sqrt{I}$ follows from $B \subseteq I$, while $x \in I$ implies $x^n \in B$, proving $I \subseteq \sqrt{B}$ and hence (b). Taking $(I, I, 1)$ proves (c). Finally, in an SFT ring every ideal has the same radical as a finitely generated ideal. Thus every closed subset of $\text{Spec } A$ is quasi-compact, equivalently $\text{Spec } A$ is Noetherian. $\square$

2.2. Quotients, lifting, and flat descent.

Lemma 2.4 (Quotients [3, Proposition 2.3]). *Every quotient of an SFT ring is SFT. More precisely, if $L \supseteq J$ and (L, B, n) is SFT data in A , then*

$$(L/J, (B + J)/J, n)$$

is SFT data in A/J .

Proof. For $x + J \in L/J$, the relation $x^n \in B$ gives $(x + J)^n \in (B + J)/J$. The image of the finite ideal B is finite. $\square$

Lemma 2.5 (Lifting across an SFT ideal) [Project result].

Let $I \subseteq J$ be ideals of A . If I is SFT in A and J/I is SFT in A/I , then J is SFT in A .

Proof. Choose data (I, C, r) . Lift a finite witness for J/I to a finitely generated ideal $B \subseteq J$, and let n be its quotient exponent. For $x \in J$ there are $b \in B$ and $i \in I$ with $x^n = b + i$. In $(b + i)^r$, every term except i^r has a factor from B , while $i^r \in C$. Therefore

$$x^{nr} \in B + C \subseteq J,$$

so $(J, B + C, nr)$ is SFT data. $\square$

Lemma 2.6 (Faithfully flat descent) [Project result].

If $A \rightarrow C$ is faithfully flat and C is SFT, then A is SFT.

Proof. Fix $I \triangleleft A$ and choose SFT data (IC, J, n) in C , with $J = (z_1, \dots, z_t)C$. Express the z_i using finitely many elements of I , and let $K \subseteq I$ be the finitely generated A -ideal generated by all those elements. Then $J \subseteq KC$. Hence $a^n \in KC$ for every $a \in I$. Faithful flatness makes $A/K \rightarrow C/KC$ injective, so $KC \cap A = K$. Thus $a^n \in K$, and (I, K, n) is SFT data. $\square$

Lemma 2.7 (Nilpotent thickening) [Project result].

If $N \triangleleft A$, $N^s = 0$, and A/N is SFT, then A is SFT.

Proof. Every prime P contains N . Lift SFT data for P/N to a finite ideal $B \subseteq P$, so $x^n \in B + N$ for $x \in P$. Then

$$x^{ns} \in (B + N)^s \subseteq B + N^s = B.$$

Every prime is SFT, and theorem 2.2 finishes the proof. $\square$

Lemma 2.8 (Uniformly radical ascent) [Project result].

Suppose $A \subseteq C$ has characteristic $p > 0$ and one integer $e \geq 0$ satisfies $c^{p^e} \in A$ for every $c \in C$. If A is SFT, then C is SFT.

Proof. For $Q \in \text{Spec } C$, put $P = Q \cap A$ and choose SFT data (P, B, n) . If $q \in Q$, then $q^{p^e} \in P$, and therefore $q^{p^e n} \in B \subseteq BC$. Thus every prime of C is SFT, and the prime criterion applies. $\square$

2.3. Finite-module invariance. The next theorem is used in the residual-normalization and formal-fibre arguments. No separability, flatness, reducedness, or domain hypothesis is present.

Lemma 2.9 (A finite-module bound; compare [8, Theorem 4.13]) [Project result].

Let $A \rightarrow C$ be a ring map, suppose C is generated by d elements as an A -module, and let (I, J, n) be SFT data in A . Then IC has the finite witness JC and exponent

$$d(n - 1) + 1.$$

Proof. Write $C = Au_1 + \dots + Au_d$. Every $x \in IC$ has an expression $x = \sum_{i=1}^d a_i u_i$ with $a_i \in I$. Put $N = d(n - 1) + 1$. In every degree- N monomial in the expansion of x^N , one a_i occurs at least n times. That monomial belongs to JC because $a_i^n \in J$. Consequently $x^N \in JC$. $\square$

Theorem 2.10 (Module-finite ascent) [Project result].

If A is SFT and C is finite as an A -module, then C is SFT.

Proof. By lemma 2.3, $\text{Spec } A$ is Noetherian. We use Noetherian induction on closed subsets. For a closed set Z , let $\mathcal{P}(Z)$ be the assertion that for every $I \subseteq A$ with $V(I) = Z$, every finite A/I -algebra is SFT. Assume $\mathcal{P}(Z')$ for each proper closed $Z' \subsetneq Z$.

First let $I = \sqrt{I}$, put $R = A/I$, and let D be a finite algebra over the reduced SFT ring R . Fix $Q \in \text{Spec } D$ and put $P = Q \cap R$. If $P \neq 0$, then $V_R(P)$ is a proper closed subset because R

is reduced. Induction makes D/PD SFT; Lemma 2.9 makes PD an SFT ideal; and Lemma 2.5 makes Q SFT.

Suppose $P = 0$. Then R is a domain. With $K = \text{Frac } R$, the algebra $D_K = D \otimes_R K$ is Artinian. Choose $q_1, \dots, q_t \in Q$ whose images generate Q_K , and put $H = (q_1, \dots, q_t)D$. The module Q/H is torsion inside the finite R -module D/H . We use the elementary fact that the entire torsion submodule of a finite module over a domain is annihilated by one nonzero scalar: choose elements whose images form a K -basis, let F be their free R -span, clear the finitely many denominators needed to put the whole module inside F , and observe that F has no torsion. Hence there is $0 \neq d \in R$ such that

$$dQ \subseteq H. \quad (2.1)$$

If d is a unit then $Q = H$. Otherwise induction makes D/dD SFT. Choose quotient data for $(Q + dD)/dD$, represented by $h_1, \dots, h_s \in Q$ with exponent e . For $x \in Q$,

$$x^e \in (h_1, \dots, h_s)D + dD.$$

The error belongs to $Q \cap dD = dQ$: indeed $d \notin Q$ and Q is prime. By (2.1) it lies in H . Therefore Q has a fixed finite witness. Every prime of D is SFT, so D is SFT.

This proves $\mathcal{P}(Z)$ for the radical ideal defining Z . For an arbitrary I with the same closed support, set $N = \sqrt{I}/I$ in A/I . For a finite A/I -algebra D , the quotient D/ND is SFT by the radical case. The ideal ND is SFT by Lemma 2.9 and is nil, hence belongs to every prime. Lifting quotient data prime by prime proves that D is SFT. The induction is complete. $\square$

Theorem 2.11 (Module-finite descent and invariance) [Project result].

Let $A \subseteq C$ be module-finite, generated by d elements as an A -module. If IC is SFT for an ideal $I \triangleleft A$, then I is SFT. Consequently

$$A \text{ is SFT} \iff C \text{ is SFT}.$$

Proof. Choose SFT data (IC, J, n) . A finite set of C -generators of $J \subseteq IC$ involves only finitely many coefficients from I ; let $K \subseteq I$ be the finite A -ideal generated by them. Then $J \subseteq KC$. For $x \in I$, put $y = x^n$. We have $yC \subseteq KC$.

Choose A -module generators $u_1, \dots, u_d$ of C and write

$$yu_i = \sum_{j=1}^d a_{ij}u_j, \quad a_{ij} \in K.$$

The determinant trick applied to $yI_d - (a_{ij})$ gives

$$y^d + c_1y^{d-1} + \dots + c_d = 0, \quad c_i \in K^i.$$

Indeed, the adjugate annihilates all module generators; its determinant annihilates the faithful A -module C , hence vanishes in A . Since $K^i \subseteq K$ for $i \geq 1$, the equation gives $x^{nd} = y^d \in K$. Thus (I, K, nd) is SFT data. Descent follows by applying this to every I , and ascent is Theorem 2.10. $\square$

Corollary 2.12 (Finite mixed extensions) [Project result].

For a module-finite inclusion $A \subseteq C$ and finite sets of variables $\mathbf{X}, \mathbf{Y}$,

$$A[[\mathbf{Y}]][\mathbf{X}] \text{ is SFT} \iff C[[\mathbf{Y}]][\mathbf{X}] \text{ is SFT}.$$

Moreover $C[[\mathbf{Y}]]$ is module-finite and integral over $A[[\mathbf{Y}]]$, and

$$\dim C[[\mathbf{Y}]] = \dim A[[\mathbf{Y}]].$$

Proof. If $c_1, \dots, c_d$ generate C over A , collect coefficients to see that they generate $C[[\mathbf{Y}]]$ over $A[[\mathbf{Y}]]$; adjoining polynomial variables preserves module-finiteness. Apply Theorem 2.11. The dimension equality follows from lying over and going up in one direction and incomparability in the other. $\square$

Proposition 2.13 (A conductor criterion) [Project result].

For an arbitrary extension $A \subseteq C$, let $(A : C) = \{a \in A : aC \subseteq A\}$. If A and $C/(A : C)$ are SFT, then C is SFT.

Proof. Put $H = (A : C)$. For $L \triangleleft C$, the ideal $I = L \cap H$ is both an A -ideal and a C -ideal. Choose A -SFT data (I, F, r) . In $(L + H)/H$, choose a finite witness represented by $g_1, \dots, g_s \in L$, with exponent n . Then

$$x^n \in (g_1, \dots, g_s)C + I \quad (x \in L).$$

Raising to the r th power, every mixed term lies in the first ideal and the pure I -term lies in FC . Thus $x^{nr} \in (g_1, \dots, g_s)C + FC \subseteq L$. $\square$

2.4. Uppers to zero.

Definition 2.14. For a domain D , an *upper to zero* is a nonzero prime $Q \subset D[X]$ with $Q \cap D = 0$. If $u = a/b \in \text{Frac } D$, then

$$Q_u = \ker(D[X] \longrightarrow \text{Frac } D, X \longmapsto u)$$

is an upper to zero: it contains $bX - a \neq 0$, is prime, and has zero contraction.

Theorem 2.15 (Residual-upper criterion) [Project result].

For every ring R , the following are equivalent.

- (i) $R[X]$ is SFT.
- (ii) R is SFT and, for every $P \in \text{Spec } R$, every upper to zero of $(R/P)[X]$ is SFT.

Proof. If $R[X]$ is SFT, quotient first by (X) and then by $P[X]$ to obtain the necessary SFT rings. Conversely, take $Q \in \text{Spec } R[X]$ and put $P = Q \cap R$. The extension $P[X]$ is SFT by the scalar-extension theorem for SFT ideals [8, Theorem 4.13]. If $Q = P[X]$ we are done. Otherwise $Q/P[X]$ is an upper to zero in $(R/P)[X]$, and is SFT by hypothesis. Apply Lemma 2.5 and then theorem 2.2. $\square$

Corollary 2.16 [Project result].

If D is a one-dimensional SFT domain, then

$$D[X] \text{ is SFT} \iff \text{every upper to zero of } D[X] \text{ is SFT}.$$

Proof. For each nonzero prime P , the domain D/P is a field, so $(D/P)[X]$ is Noetherian. The zero residue is the only case left in Theorem 2.15. $\square$

3. POLYNOMIAL EXTENSIONS AND EQUICHARACTERISTIC COUNTEREXAMPLES

The polynomial counterexamples used throughout the log are due to Tran et al. [22, Theorems 4.1 and 5.1]. We reproduce their algebraic construction for reference. The exact dimension calculations and the later strengthened properties are project results. We first isolate a positive-characteristic mechanism based on a Frobenius-stable filtration, then construct the required filtrations, replace Frobenius by a roots-of-unity norm in equicharacteristic zero. We retain the notation *SFT data* (I, B, e) and the elementary permanence results from the preceding section.

Theorem 3.1 (Equicharacteristic polynomial counterexamples) [Project result].

The following statements hold.

- (a) For every prime p , there is a one-dimensional local SFT domain R_p of characteristic p such that $R_p[X]$ is not SFT. Its bad prime may be chosen to be an upper to zero, and $\dim R_p[X] = 3$.
- (b) There is a one-dimensional local SFT domain R_0 containing $\mathbb{Q}$ such that $R_0[X]$ has a non-SFT upper to zero and $\dim R_0[X] = 3$.

In both cases the coefficient domain is non-Noetherian, and adjoining any positive finite number of polynomial indeterminates produces a non-SFT ring.

The proof occupies the rest of the section.

3.1. A Frobenius–Rees criterion. The filtered Rees localization below is the construction used in the cited counterexample paper. Fix fields $k \subseteq K$ of characteristic $p > 0$. Let $\mathcal{V} = (V_n)_{n \geq 0}$ be k -vector subspaces of K satisfying

- (F1) $V_0 = k$, the sequence is increasing, and $\bigcup_{n \geq 0} V_n = K$;
- (F2) $V_n V_m \subseteq V_{n+m}$ for all $m, n \geq 0$;
- (F3) $v^p \in V_n$ for every $v \in V_n$.

Here $V_n V_m$ is the k -span of the products vw with $v \in V_n$ and $w \in V_m$. Let c be transcendental over K and set

$$A = \bigoplus_{n \geq 0} c^n V_n \subseteq K[c], \quad \mathfrak{m} = A_+ = \bigoplus_{n \geq 1} c^n V_n, \quad R = A_{\mathfrak{m}}, \quad M = \mathfrak{m}R. \quad (3.1)$$

Multiplicativity makes A a subring of the domain $K[c]$. Taking constant coefficients gives $A/\mathfrak{m} \cong k$, so $\mathfrak{m}$ is maximal and R is a local domain with maximal ideal M . Moreover $1 \in V_n$ for every n , and therefore $c^n \in A$.

Lemma 3.2 (Formal coefficient bounds) [Project result].

Set

$$\mathcal{S}_{\mathcal{V}} = \left\{ \sum_{n \geq 0} c^n a_n \in K[[c]] : a_n \in V_n \right\}.$$

Then $\mathcal{S}_{\mathcal{V}}$ is a subring of $K[[c]]$, and the canonical injection

$$R \longrightarrow K[c]_{(c)} \longrightarrow K[[c]]$$

has image in $\mathcal{S}_{\mathcal{V}}$. Thus every $r \in R$ has a unique expansion

$$r = \sum_{n \geq 0} c^n r_n, \quad r_n \in V_n. \quad (3.2)$$

Proof. The coefficient of c^n in a product of two elements of $\mathcal{S}_{\mathcal{V}}$ is the finite sum

$$\sum_{i=0}^n a_i b_{n-i} \in \sum_{i=0}^n V_i V_{n-i} \subseteq V_n.$$

Hence $\mathcal{S}_{\mathcal{V}}$ is a ring. If $s \in A \setminus \mathfrak{m}$, its constant coefficient s_0 belongs to $k^\times$. After dividing by s_0 , write

$$s_0^{-1} s = 1 + \sum_{i=1}^N c^i s_i, \quad s_i \in V_i.$$

Its inverse in $K[[c]]$ is $\sum_{n \geq 0} c^n t_n$, where $t_0 = 1$ and

$$t_n = - \sum_{i=1}^{\min\{n, N\}} s_i t_{n-i}.$$

Induction on n and (F2) give $t_n \in V_n$. Consequently s^{-1} , and then every fraction a/s defining an element of R , lies in $\mathcal{S}_{\mathcal{V}}$. Since $s(0) \neq 0$, the fraction lies in $K[c]_{(c)}$, whose canonical map to $K[[c]]$ is injective. This also proves uniqueness. $\square$

Lemma 3.3 (Uniform powers and cofinal divisibility) [Project result].

The following statements hold.

- (a) $z^p \in cR$ for every $z \in M$.

(b) For every $0 \neq a \in M$, there is $D \geq 1$ such that $c^D \in aR$.

Proof. For (a), first let $b = \sum_{n=1}^N c^n b_n \in \mathfrak{m}$. Frobenius and (F3) give

$$\frac{b^p}{c} = \sum_{n=1}^N c^{pn-1} b_n^p \in A, \quad (3.3)$$

because $b_n^p \in V_n \subseteq V_{pn-1}$. If $z = b/s \in M$ with $s \notin \mathfrak{m}$, then $z^p = c((b^p/c)/s^p) \in cR$.

For (b), write $a = b/t$ with $b \in \mathfrak{m}$ and $t \in A \setminus \mathfrak{m}$. Since t is a unit, $aR = bR$. Write

$$b = \sum_{j=\nu}^N c^j b_j, \quad \nu \geq 1, \quad b_\nu \neq 0, \quad b_j \in V_j.$$

By exhaustivity choose $s \geq 0$ such that $b_\nu^{-1} \in V_s$. Choose $r \geq 0$ so large that

$$(j - \nu)p^r \geq j + s \quad \text{for every } j > \nu \text{ with } b_j \neq 0. \quad (3.4)$$

Repeated use of (F3) gives $b_\nu^{-p^r} \in V_s$, and hence $\lambda = c^s b_\nu^{-p^r} \in A$. For every higher nonzero coefficient,

$$\left(\frac{b_j}{b_\nu}\right)^{p^r} = b_j^{p^r} b_\nu^{-p^r} \in V_j V_s \subseteq V_{j+s} \subseteq V_{(j-\nu)p^r}.$$

Applying Frobenius to b and factoring the term of least c -degree yields

$$\lambda b^{p^r} = c^{\nu p^r + s} \left(1 + \sum_{j > \nu} c^{(j-\nu)p^r} \left(\frac{b_j}{b_\nu}\right)^{p^r} \right) = c^{\nu p^r + s} (1 + w), \quad (3.5)$$

where $w \in \mathfrak{m}$. Thus $1 + w$ is a unit and, with $D = \nu p^r + s$,

$$c^D = (1 + w)^{-1} \lambda b^{p^r} \in bR = aR.$$

□

Theorem 3.4 (The coefficient domain) [Project result].

The ring R in (3.1) is a one-dimensional local SFT domain. More precisely, every nonzero proper ideal I has SFT data $(I, (a), pD)$ for some $0 \neq a \in I$ and $D \geq 1$, and

$$\text{Spec } R = \{(0), M\}. \quad (3.6)$$

Proof. Let I be a nonzero proper ideal and choose $0 \neq a \in I$. Since R is local, $I \subseteq M$. By Lemma 3.3(b), $c^D \in aR$ for some D . For every $z \in I$, part (a) gives

$$z^{pD} = (z^p)^D \in c^D R \subseteq aR.$$

Thus $(I, (a), pD)$ is SFT data. The zero and unit ideals are immediate.

If P is a nonzero prime, choose $0 \neq a \in P$. Cofinal divisibility gives $c^D \in aR \subseteq P$, so $c \in P$. For $z \in M$, part (a) gives $z^p \in cR \subseteq P$, and therefore $z \in P$. Thus $P = M$. Since $0 \neq c \in M$, the two primes in (3.6) are distinct and $\dim R = 1$. □

Fix $u \in V_1$ and impose the escape condition

$$k[u] \not\subseteq V_E \quad (E \geq 0). \quad (3.7)$$

The filtration axioms give

$$u^j \in V_j \quad (j \geq 0), \quad u^{p^n} \in V_1 \quad (n \geq 0). \quad (3.8)$$

Lemma 3.5 (Division by the evaluation equation) [Project result].

Let $h \in R[X]$ have degree at most $d \geq 1$ and satisfy $h(u) = 0$ in $K(c)$. Writing $h = (X - u)\tilde{h}$ in $K(c)[X]$, the coefficient of c^n in each X -coefficient of $\tilde{h}$ belongs to V_{n+d-1} .

Proof. Write $h = \sum_{j=0}^d h_j X^j$. Since $h(u) = 0$,

$$h(X) = h(X) - h(u) = (X - u) \sum_{j=1}^d h_j \sum_{t=0}^{j-1} X^{j-1-t} u^t. \quad (3.9)$$

By Lemma 3.2, the c^n -coefficient of h_j lies in V_n , while (3.8) gives $u^t \in V_t$. Therefore each required coefficient lies in $\sum_{t=0}^{d-1} V_n V_t \subseteq V_{n+d-1}$. $\square$

Theorem 3.6 (A non-SFT upper to zero) [Project result].

Under (3.7),

$$Q_u = \ker(R[X] \longrightarrow K(c), X \longmapsto u)$$

is a non-SFT upper to zero. Consequently $R[X]$ is not SFT.

Proof. If $q \in K$, choose n with $q \in V_n$. Since $c^n q, c^n \in A$, $q = (c^n q)/c^n \in \text{Frac}(R)$. Together with $R \subseteq K[c]_{(c)}$, this proves

$$\text{Frac}(R) = K(c). \quad (3.10)$$

The quotient $R[X]/Q_u \cong R[u]$ is a domain, and evaluation is injective on R . Hence Q_u is prime and $Q_u \cap R = (0)$. It is nonzero because

$$g_n := c(X - u)^{p^n} = cX^{p^n} - cu^{p^n} \in Q_u \quad (n \geq 0); \quad (3.11)$$

indeed $u^{p^n} \in V_1$.

Suppose that Q_u had SFT data (Q_u, J, e) , with $J = (h_1, \dots, h_t)$. The ideal J is nonzero because otherwise $g_0^e = 0$ in the domain $R[X]$. Discard zero generators and put $d = \max_i \deg h_i \geq 1$. Choose r with $q = p^r \geq e$. Enlargement of the SFT exponent gives $f^q \in J$ for every $f \in Q_u$, and hence

$$g_n^q = \sum_{i=1}^t a_{i,n} h_i, \quad a_{i,n} \in R[X]. \quad (3.12)$$

Write $h_i = (X - u)\tilde{h}_i$ by Lemma 3.5. Substitution and cancellation of $X - u$ give

$$c^q (X - u)^{p^{n+r}-1} = \sum_{i=1}^t a_{i,n} \tilde{h}_i. \quad (3.13)$$

Write

$$a_{i,n} = \sum_{\alpha} A_{i,\alpha}(c) X^{\alpha}, \quad \tilde{h}_i = \sum_{\beta} H_{i,\beta}(c) X^{\beta}.$$

Then $[c^s]A_{i,\alpha} \in V_s$ and $[c^\ell]H_{i,\beta} \in V_{\ell+d-1} \subseteq V_{\ell+d}$. For fixed X -degree m ,

$$[c^q X^m](a_{i,n} \tilde{h}_i) = \sum_{\alpha+\beta=m} \sum_{s+\ell=q} [c^s]A_{i,\alpha} [c^\ell]H_{i,\beta}. \quad (3.14)$$

Both sums are finite, and every summand lies in $V_s V_{\ell+d} \subseteq V_{q+d}$. Comparing the coefficient of c^q in (3.13) therefore puts every coefficient of $(X - u)^{p^{n+r}-1}$ in the fixed space V_{q+d} . But

$$(X - u)^{p^N-1} = \frac{X^{p^N} - u^{p^N}}{X - u} = \sum_{j=0}^{p^N-1} u^j X^{p^N-1-j}. \quad (3.15)$$

Thus $1, u, \dots, u^{p^{n+r}-1} \in V_{q+d}$ for every n , contradicting (3.7). Hence Q_u is not SFT. $\square$

3.2. Weighted filtrations in prime characteristic. For $L \geq 0$, put $K^{p^L} = \{z^{p^L} : z \in K\}$.

Lemma 3.7 (Standard Frobenius-degree lemma; compare [19, Tag 09HF]). *For $u \in K$, the following conditions are equivalent:*

- (i) $u \notin K^p$;
- (ii) $[K^{p^L}(u) : K^{p^L}] = p^L$ for every $L \geq 0$.

When they hold, $1, u, \dots, u^{p^L-1}$ is a basis of $K^{p^L}(u)$ over K^{p^L} .

Proof. Condition (ii) with $L = 1$ implies (i). Conversely, let $u \notin K^p$. The element u is purely inseparable over K^{p^L} and $u^{p^L} \in K^{p^L}$. If its degree were p^r for some $r < L$, then $u^{p^r} \in K^{p^L}$, say $u^{p^r} = z^{p^L}$. Injectivity of Frobenius would give $u = z^{p^{L-r}} \in K^p$, a contradiction. Hence the degree is p^L . The displayed powers span and their number is the degree, so they form a basis. $\square$

Proposition 3.8 (Construction from an enumeration and weights) [Project result].

Let k be a countable perfect field of characteristic p , let K be a countable imperfect extension field, and choose $u \in K \setminus K^p$. Enumerate

$$K^\times = \{\xi_1, \xi_2, \dots\}, \quad \xi_1 = u,$$

and choose positive integer weights w_i such that

$$w_1 = 1, \quad \#\{i : w_i \leq E\} < \infty \quad (E \geq 0). \quad (3.16)$$

For $n \geq 0$, let V_n be the k -span of all monomials

$$\prod_{\ell=1}^r \xi_{i_\ell}^{p^{e_\ell}}, \quad r \geq 0, \quad e_\ell \geq 0, \quad \sum_{\ell=1}^r w_{i_\ell} \leq n, \quad (3.17)$$

with repeated factors allowed and the empty product equal to 1. Then (F1)–(F3) and (3.7) hold, and $u \in V_1$.

Proof. Positive weights give $V_0 = k$ and monotonicity. Multiplying monomials adds their weights, so $V_n V_m \subseteq V_{n+m}$. Frobenius raises every p^{e_ℓ} to $p^{e_\ell+1}$ without changing the weights; coefficients remain in k . Every nonzero element of K is some ξ_i and belongs to V_{w_i} , proving exhaustivity. Finally $u \in V_1$.

Fix E and let

$$F_L = K^{p^L}, \quad N_E = \#\{i : w_i \leq E\}.$$

Since k is perfect, $k \subseteq F_L$. In a spanning monomial for V_E , every factor with Frobenius exponent at least L lies in F_L . The remaining part has at most E factors, each selected from a set of at most $N_E L$ elements. Therefore

$$\dim_{F_L}(F_L V_E) \leq \sum_{r=0}^E (N_E L)^r. \quad (3.18)$$

For fixed E this is eventually less than p^L . By lemma 3.7, the p^L powers $1, u, \dots, u^{p^L-1}$ are independent over F_L . At least one lies outside $F_L V_E$, and hence outside V_E . This proves (3.7). $\square$

Corollary 3.9 (Prime-characteristic family) [Project result].

Under the hypotheses of Proposition 3.8, the Rees localization (3.1) is a one-dimensional local SFT domain, whereas $R[X]$ is not SFT and has a non-SFT upper to zero. In particular, for every prime p one may take

$$k = \mathbf{F}_p, \quad K = \mathbf{F}_p(u), \quad w_i = i$$

after enumerating $K^\times$ with $\xi_1 = u$. Denote this explicit ring by R_p , its maximal ideal by M_p , and its bad upper by Q_p .

Proof. Apply Theorems 3.4 and 3.6. For $K = \mathbf{F}_p(u)$, the element u is transcendental and hence does not belong to K^p . $\square$

3.3. The equicharacteristic-zero construction. In characteristic zero, Frobenius is unavailable. An exponentially indexed word filtration gives a cubic containment for the maximal ideal, while a roots-of-unity norm replaces the Frobenius calculation in the divisibility argument.

Theorem 3.10 (Equicharacteristic-zero counterexample) [Project result].

There is a one-dimensional local SFT domain R_0 containing $\mathbb{Q}$ such that $R_0[X]$ has a non-SFT upper to zero.

Proof. Let $k = \overline{\mathbb{Q}}$, let u be transcendental over k , and put $K = k(u)$. Enumerate $K^\times = \{\xi_1, \xi_2, \dots\}$ with $\xi_1 = 1$. Use the labelled alphabet

$$\{a_j : j \geq 0\} \amalg \{x_i : i \geq 1\}, \quad a_j \mapsto u^{2^j}, \quad x_i \mapsto \xi_i,$$

with weights $w(a_j) = 1$ and $w(x_i) = i$. Labels are retained even when their images in K coincide. Put $B_n = 2^n - 1$, and define

$$V_n = \text{Span}_k\{\text{images of words of total weight at most } B_n\} \subseteq K, \quad (3.19)$$

where the empty word is allowed.

The spaces are increasing and exhaustive, with $V_0 = k$. Concatenation and the inequalities

$$B_n + B_m \leq B_{n+m}, \quad B_i + B_j + B_\ell \leq B_{i+j+\ell-1} \quad (i, j, \ell \geq 1)$$

give

$$V_n V_m \subseteq V_{n+m}, \quad V_i V_j V_\ell \subseteq V_{i+j+\ell-1} \quad (i, j, \ell \geq 1). \quad (3.20)$$

The labels a_j also give

$$u^{2^j} \in V_1 \quad (j \geq 0), \quad u^t \in V_t \quad (t \geq 0). \quad (3.21)$$

For the second assertion, represent t as a sum of powers of two; the resulting word has at most $t \leq B_t$ letters when $t \geq 1$.

We claim that

$$k[u] \not\subseteq V_E \quad (E \geq 0). \quad (3.22)$$

Fix E and put $B = B_E$. A word of weight at most B has image $h(u)u^s$, where h belongs to a finite set of products of $\xi_1, \dots, \xi_B$ and s is a sum of at most B powers of two. Let S_B be the set of those sums. Up to L , each summand has at most $1 + \lfloor \log_2 L \rfloor$ choices, so

$$\#(S_B \cap [0, L]) = O_B((\log L)^B).$$

Clear one common denominator for the finite set of possible h . There are $0 \neq \Delta \in k[u]$ and a finite $F \subseteq \mathbb{N}$ such that

$$\text{supp}(\Delta f) \subseteq F + S_B \quad (f \in V_E). \quad (3.23)$$

If $k[u] \subseteq V_E$, then for any $r \in \text{supp}(\Delta)$ the polynomials Δu^N would imply $r + \mathbb{N} \subseteq F + S_B$. The left side has linear growth, whereas the right side has polylogarithmic growth. This proves (3.22).

Set

$$A_0 = \bigoplus_{n \geq 0} c^n V_n \subseteq K[c], \quad \mathfrak{m}_0 = (A_0)_+, \quad R_0 = (A_0)_{\mathfrak{m}_0}, \quad M_0 = \mathfrak{m}_0 R_0. \quad (3.24)$$

The first inclusion in (3.20) makes A_0 a graded domain, and $A_0/\mathfrak{m}_0 \cong k$. Hence R_0 is local. The proof of Lemma 3.2 uses only multiplicativity and applies verbatim: every $r \in R_0$ has a unique expansion

$$r = \sum_{n \geq 0} c^n r_n \in K[[c]], \quad r_n \in V_n. \quad (3.25)$$

The triple-product estimate gives

$$z^3 \in cR_0 \quad (z \in M_0). \quad (3.26)$$

Indeed, if $b = \sum_{i \geq 1} c^i b_i \in \mathfrak{m}_0$, then the coefficient of c^r in b^3/c is a sum of products $b_i b_j b_\ell \in V_{i+j+\ell-1} = V_r$. Thus $b^3/c \in A_0$, and localization proves the assertion.

We next prove cofinal divisibility. Let $0 \neq a \in M_0$ and write $a = b/t$, where $t \notin \mathfrak{m}_0$ and

$$b = \sum_{j=\nu}^L c^j b_j \in \mathfrak{m}_0, \quad b_\nu \neq 0.$$

Since t is a unit, $aR_0 = bR_0$. Choose i_0 with $\xi_{i_0} = b_\nu^{-1}$ and put $w_0 = i_0$. Choose q so large that

$$q(B_L + w_0) \leq B_q = 2^q - 1. \quad (3.27)$$

The field k contains all q th roots of unity. Let $\mu_q = \{\zeta \in k : \zeta^q = 1\}$ and form

$$\mathcal{N}_q(b) = \prod_{\zeta \in \mu_q} b(\zeta c).$$

After factoring the least term, invariance under $c \mapsto \eta c$ for $\eta \in \mu_q$ gives

$$\mathcal{N}_q(b) = \varepsilon c^{\nu q} b_\nu^q \left(1 + \sum_{r \geq 1} \theta_r c^{r q} \right), \quad \varepsilon \in k^\times, \quad (3.28)$$

with only finitely many nonzero θ_r . Each ratio b_j/b_ν is a k -linear combination of words of weight at most $B_L + w_0$. Therefore every product of at most q such ratios has weight at most $q(B_L + w_0) \leq B_q$, and hence $\theta_r \in V_q \subseteq V_{rq}$. The same estimate gives $b_\nu^{-q} \in V_q$. Thus

$$\lambda = c^q b_\nu^{-q} \in A_0, \quad \omega = \sum_{r \geq 1} \theta_r c^{r q} \in \mathfrak{m}_0.$$

Every $b(\zeta c)$ belongs to A_0 , and the factor corresponding to $\zeta = 1$ is b . Hence $\mathcal{N}_q(b) \in bA_0$. Equation (3.28) now gives

$$c^{\nu q + q} = \varepsilon^{-1} (1 + \omega)^{-1} \lambda \mathcal{N}_q(b) \in bR_0 = aR_0. \quad (3.29)$$

Let I be a nonzero proper ideal of R_0 and choose $0 \neq a \in I$. Equation (3.29) gives $c^D \in aR_0$ for some D , and (3.26) gives $z^{3D} \in c^D R_0 \subseteq aR_0$ for every $z \in I$. Thus $(I, (a), 3D)$ is SFT data. The same two estimates show that every nonzero prime contains c and then M_0 . Therefore

$$\text{Spec } R_0 = \{(0), M_0\}, \quad \dim R_0 = 1. \quad (3.30)$$

The domain contains k , and its residue field is k , so it is equicharacteristic zero.

Exhaustivity gives $\text{Frac}(R_0) = K(c)$. Define

$$Q_0 = \ker(R_0[X] \longrightarrow K(c), X \mapsto u).$$

This is a prime with zero contraction, and it is nonzero because, for $m = 2^n$,

$$g_n = c(X^m - u^m) \in Q_0; \quad (3.31)$$

indeed $u^m \in V_1$. Thus Q_0 is an upper to zero.

Suppose that Q_0 had SFT data (Q_0, J, e) , where $J = (h_1, \dots, h_t)$. As before, $J \neq 0$ and $d = \max_i \deg h_i \geq 1$. Fix $N \geq e$. The proof of Lemma 3.5, now using (3.21), writes $h_i = (X - u)\tilde{h}_i$ with every c^ℓ -coefficient of $\tilde{h}_i$ in $V_{\ell+d-1}$. Since $g_n^N \in J$, cancellation gives

$$c^N \frac{(X^m - u^m)^N}{X - u} = \sum_{i=1}^t a_{i,n} \tilde{h}_i, \quad a_{i,n} \in R_0[X]. \quad (3.32)$$

The coefficient of c^N is a finite convolution. Every term on the right belongs to $V_s V_{\ell+d-1} \subseteq V_{N+d}$, so every coefficient of

$$G_{m,N} = \frac{(X^m - u^m)^N}{X - u}$$

lies in the fixed space V_{N+d} .

Use

$$G_{m,N} = (X^m - u^m)^{N-1} \sum_{j=0}^{m-1} X^{m-1-j} u^j.$$

For $0 \leq t < mN$, put $s = \lfloor t/m \rfloor + 1$. Exactly one term in the second factor has the required residue class modulo m , and multiplication gives

$$[X^t]G_{m,N} = (-1)^{N-s} \binom{N-1}{s-1} u^{mN-1-t}. \quad (3.33)$$

The scalar is nonzero in characteristic zero. Consequently $1, u, \dots, u^{N2^n-1} \in V_{N+d}$ for every n , contradicting (3.22). Hence Q_0 is not SFT. $\square$

3.4. Prime chains, dimensions, and normality for the direct examples.

Proposition 3.11 (Location of the bad upper and exact dimension) [Project result].

Let (D, N, Q) be either (R, M, Q_u) from Theorems 3.4 and 3.6, or (R_0, M_0, Q_0) from Theorem 3.10. Then

$$(0) \subsetneq Q \subsetneq N[X] \subsetneq (N, X) \quad (3.34)$$

is a strict chain of prime ideals, and $\dim D[X] = 3$.

Proof. In positive characteristic, escape implies that u is transcendental over k : if it were algebraic, all its powers would belong to the span of finitely many powers and hence to one filtration level. In characteristic zero, transcendence was part of the construction.

Let $h = \sum_j r_j X^j \in Q$. The formal expansion of each r_j has constant coefficient in the residue field k . Taking the constant c -coefficient of $h(u) = 0$ gives

$$\sum_j \bar{r}_j u^j = 0, \quad \bar{r}_j \in k.$$

Transcendence forces every $\bar{r}_j = 0$, so $Q \subseteq N[X]$. The inclusion is strict because $c \in N[X]$ but $Q \cap D = (0)$. The remaining inclusions are strict because $Q \neq 0$ and $X \in (N, X) \setminus N[X]$. All four ideals are prime; the last two quotients are $k[X]$ and k .

The chain gives $\dim D[X] \geq 3$. Conversely, $\text{Spec } D = \{(0), N\}$. For any fixed contraction P in a prime chain of $D[X]$, quotient by $P[X]$ and localize the domain D/P at its nonzero elements. The corresponding part of the chain embeds in the one-dimensional PID $\text{Frac}(D/P)[X]$, so at most two primes have contraction P . There are only two possible contractions, hence at most four primes in the chain and $\dim D[X] \leq 3$. $\square$

Proposition 3.12 (Failure of normality) [Project result].

Every positive-characteristic domain furnished by Corollary 3.9, and the domain R_0 , is not integrally closed. Consequently none is a Prüfer domain, and none has every prime factor integrally closed.

Proof. In positive characteristic choose $j \geq 1$ with $u^j \notin V_1$ and set $z = cu^j \in K(c) = \text{Frac}(R)$. The formal coefficient bound shows $z \notin R$. Since $u^j \in V_j$, choose N with $p^N \geq j$. Frobenius stability gives

$$u^{jp^N} \in V_j \subseteq V_{p^N},$$

and hence $z^{p^N} \in c^{p^N} V_{p^N} \subseteq R$. Thus z is integral over R but does not belong to it.

For R_0 , (3.19) gives $V_1 = \text{Span}_k(\{1\} \cup \{u^{2^j} : j \geq 0\})$, so $u^3 \notin V_1$ and $cu^3 \notin R_0$. On the other hand, $u^6 = (u^2)^3 \in V_1^3 \subseteq V_2$, and hence $(cu^3)^2 = c^2 u^6 \in R_0$. This is again a missing integral element. Every Prüfer domain is integrally closed, and the zero prime gives the last assertion. $\square$

3.5. Positive polynomial classes and the normalization obstruction. The counterexamples above are nonnormal. The residual-upper criterion and finite-module invariance give complementary positive results. Recall that a domain E is $N-1$ if its normalization in $\text{Frac } E$ is a finite E -module [19, Tag 032F], and that D is *residually $N-1$* if D/P is $N-1$ for every $P \in \text{Spec } D$.

Proposition 3.13 (Park and Barucci–Dobbs–Mulay [18][4, Theorem 4.1]). *The following domains are polynomial-stable SFT domains.*

- (a) *every one-dimensional normal SFT domain;*
- (b) *every normal SFT going-down domain.*

Proof. Park proves that an SFT domain D has $D[X]$ SFT whenever every residue domain D/P is integrally closed [18]. In (a), $D/(0) = D$ is normal and every nonzero prime quotient is a field. In (b), every factor domain of an integrally closed going-down domain is integrally closed [4, Theorem 4.1]. Park’s theorem applies in both cases. $\square$

Theorem 3.14 (Residual $N-1$ criterion) [Project result].

Let D be a finite-dimensional residually $N-1$ SFT domain. Then $D[X]$ is SFT.

Proof. We prove simultaneously, by induction on $\dim(D/P)$, that $(D/P)[X]$ is SFT for every $P \in \text{Spec } D$. A zero-dimensional residue domain is a field. For the induction step put $E = D/P$, and let B be the normalization of E in $\text{Frac } E$. The $N-1$ hypothesis makes B finite over E ; hence B is SFT by Theorem 2.10.

We verify the residual-upper criterion for $B[X]$. At the zero prime, Park’s generic-upper result for a normal SFT domain says that every upper to zero of $B[X]$ is SFT [18]. Let $0 \neq Q \in \text{Spec } B$ and set $q = Q \cap E$. Then $q \neq 0$: for $0 \neq b = a/c \in Q \subseteq \text{Frac } E$, one has $0 \neq a = bc \in Q \cap E$. Consequently $\dim(E/q) < \dim E$, and induction makes $(E/q)[X]$ SFT. The domain B/Q is finite over E/q , so Theorem 2.10 makes $(B/Q)[X]$ SFT. Thus all residue-domain uppers required by Theorem 2.15 are SFT, and $B[X]$ is SFT. Finally $E[X] \subseteq B[X]$ is module-finite, so Theorem 2.11 descends SFT to $E[X]$. Taking $P = 0$ proves the theorem. $\square$

Corollary 3.15 (Finite-normalization obstruction) [Project result].

Let D be a finite-dimensional normal SFT domain. If the normalization of D/P in its own fraction field is finite for every nonzero prime P , then $D[X]$ is SFT. Consequently, a normal finite-dimensional counterexample must have a nonzero factor with nonfinite normalization. In dimension two, finite normalization of every height-one factor suffices for polynomial stability.

Proof. The factor at $P = 0$ is already normal, while a maximal-prime factor is a field. Hence the hypotheses make D residually $N-1$, and Theorem 3.14 applies. The obstruction statement is the contrapositive. In dimension two the only remaining nonzero, nonmaximal primes have height one. $\square$

Corollary 3.16 (Every bad residue in minimal dimension) [Project result].

Suppose d is minimal among the dimensions of finite-dimensional normal SFT domains D for which $D[X]$ is not SFT. If a prime P supports a non-SFT upper to zero in $(D/P)[X]$, then $P \neq 0$ and D/P is not $N-1$.

Proof. Park’s generic-upper result for the normal SFT domain D rules out $P = 0$. Put $E = D/P$ and suppose its normalization B were finite. Then B is a normal SFT domain by Theorem 2.10, and $\dim B = \dim E < d$. The given upper makes $E[X]$ non-SFT. If $B[X]$ were SFT, module-finite descent along $E[X] \subseteq B[X]$ would make $E[X]$ SFT. Hence $B[X]$ is non-SFT, contradicting minimality of d . $\square$

3.6. Consequences and the formal-series boundary. Let S denote either R_p or R_0 , and let $\mathcal{Q} \subseteq S[X]$ denote the corresponding displayed non-SFT prime.

Corollary 3.17 (Polynomial consequences) [Project result].

The following statements hold.

- (a) *The prime $\mathcal{Q}$ is not finitely generated.*
- (b) *The coefficient domain S is non-Noetherian.*
- (c) *$S[X_1, \dots, X_m]$ is not SFT for every $m \geq 1$.*
- (d) *Each of these non-SFT polynomial rings is also not VSFT.*

Proof. A finitely generated ideal is SFT with exponent 1, proving (a). If S were Noetherian, the Hilbert basis theorem would make $S[X]$ Noetherian and hence SFT, proving (b). For $m \geq 2$, quotient $S[X_1, \dots, X_m]$ by $(X_2, \dots, X_m)$ to obtain the non-SFT ring $S[X_1]$; quotient permanence proves (c). Finally VSFT implies SFT, so (d) is immediate. $\square$

In both cases the bad prime is not extended from the coefficient ring:

$$(Q_p \cap R_p)R_p[X] = (0) \neq Q_p, \quad (Q_0 \cap R_0)R_0[X] = (0) \neq Q_0, \quad (3.35)$$

Thus permanence of individual extended SFT ideals does not control these primes.

Corollary 3.18 (Hybrid extensions) [Project result].

For every $s \geq 1$, both

$$S[[Y_1, \dots, Y_s]][X] \quad \text{and} \quad (S[X])[[Y_1, \dots, Y_s]]$$

are non-SFT. There is a natural strict inclusion

$$S[[Y_1, \dots, Y_s]][X] \subsetneq (S[X])[[Y_1, \dots, Y_s]]. \quad (3.36)$$

Proof. Quotienting either displayed ring by $(Y_1, \dots, Y_s)$ gives $S[X]$. Hence neither can be SFT. An element of the left side has a uniform finite bound on its degree in X . The series

$$\sum_{n \geq 0} X^n Y_1^n$$

belongs to the right side and has no such bound, proving strictness. $\square$

Arnold's formal-power-series dimension theorem [2, Theorem 1], applied to the non-SFT coefficient ring $S[X]$, gives

$$\dim(S[X])[[Y_1, \dots, Y_s]] = \infty \quad (s \geq 1). \quad (3.37)$$

For $s > 1$, the claim follows from the one-variable case by quotienting by $Y_2, \dots, Y_s$. The later Toan–Kang theorem [20, 21] strengthens this: already $(S[X])[[Y_1]]$ contains a prime chain of cardinality at least $2^{\aleph_1}$; see theorem 6.14.

Remark 3.19 (What the construction does not say about formal series). The rings $(S[X])[[Y]]$ and $S[[Y]][X]$ must not be identified. More importantly, none of the polynomial evaluation maps used above extends in general to a homomorphism $S[[X]] \rightarrow \text{Frac}(S)$ by substituting $X = u$: an arbitrary formal series need not have a meaningful value at u . The preceding *polynomial-evaluation* proofs therefore do not determine the SFT property or dimension of the particular formal rings. Those rings are nevertheless settled later by a different coefficient-block argument: Theorem 4.14 proves the direct non-SFT and non- t -SFT statements, and Theorem 6.12 proves the cardinal prime-chain bounds. The distinction of methods is essential; the non-SFT result for $S[[Y]][X]$ alone does not decide $S[[Y]]$.

Likewise, the SFT data constructed for R_p and R_0 are elementwise data. They do not assert containment of a full ideal power and therefore do not by themselves prove that the coefficient domains are VSFT. No conclusion about the t -SFT property of these coefficient or polynomial

domains is used here. The only VSFT conclusion forced by this section is that the non-SFT polynomial and hybrid rings cannot be VSFT.

The following table summarizes the two project examples.

Characteristic	Coefficient ring	$\dim S$	Bad-prime contraction	$\dim S[X]$
$p > 0$	R_p	1	$Q_p \cap R_p = (0)$	3
equicharacteristic 0	R_0	1	$Q_0 \cap R_0 = (0)$	3

This proves Theorem 3.1.

4. THE t -SFT AND VSFT VARIANTS

This section records what the constructions prove for the t -SFT and very-strong-finite-type variants. We use the following parts of the constructions. For $D = R_p$ or R_0 , the elements c, u , the coefficient field K , and the increasing filtration $(V_n)_{n \geq 0}$ have the properties

$$u^j \in V_j, \quad c^n V_n \subseteq D, \quad k[u] \not\subseteq V_E \quad (E \geq 0), \quad (4.1)$$

where $k = \mathbb{F}_p$ for R_p and $k = \overline{\mathbb{Q}}$ for R_0 . Every element of D , viewed in $K[[c]]$, has a unique filtered expansion

$$r = \sum_{n \geq 0} c^n r_n, \quad r_n \in V_n. \quad (4.2)$$

In particular, for $a \in K$,

$$c^E a \in D \implies a \in V_E. \quad (4.3)$$

For R_p , one also has $u^{p^j} \in V_1$ and

$$z^p \in cR_p \quad (z \in M_p). \quad (4.4)$$

For R_0 , the filtration satisfies

$$V_i V_j V_\ell \subseteq V_{i+j+\ell-1} \quad (i, j, \ell \geq 1), \quad (4.5)$$

and every $0 \neq a \in M_0$ satisfies $c^D \in aR_0$ for some $D \geq 1$. These are recalled here to make clear exactly which established properties enter the arguments below.

4.1. Definitions and general implications.

Definition 4.1. We use the t -SFT terminology of Kang–Park [14]. Let D be a domain with quotient field L . For every nonzero fractional ideal I of D , set

$$I^{-1} = (D : I) = \{x \in L : xI \subseteq D\}, \quad I_v = (I^{-1})^{-1},$$

and

$$I_t = \bigcup \{J_v : J \subseteq I \text{ is a nonzero finitely generated ideal}\}.$$

A nonzero integral ideal I is t -SFT if there are a finitely generated ideal $B \subseteq I$ and an integer $e \geq 1$ such that

$$x^e \in B_v \quad (x \in I_t).$$

The domain D is a t -SFT domain if every nonzero ideal of D is t -SFT.

Definition 4.2. We use the VSFT terminology of Coykendall–Dutta [8, Definitions 2.3–2.4]. An ideal I of a ring A is VSFT if there are a finitely generated ideal $B \subseteq I$ and an integer $n \geq 1$ such that

$$I^n \subseteq B.$$

The ring A is VSFT if every ideal is VSFT. In comparison, ordinary SFT data require only $x^n \in B$ for each individual $x \in I$. Hence

$$A \text{ VSFT} \implies A \text{ SFT}. \quad (4.6)$$

The normal case of polynomial t -SFT ascent is already known.

Theorem 4.3 (Kang–Park [14, Theorem 2.4]). *If D is an integrally closed domain, then*

$$D \text{ is } t\text{-SFT} \iff D[X] \text{ is } t\text{-SFT}.$$

We shall also use Kang and Park’s prime- t -ideal criterion [14, Proposition 2.1]: a domain is t -SFT if and only if each of its nonzero prime t -ideals is t -SFT. It gives the following useful implication.

Lemma 4.4 (Consequence of Kang–Park [14, Proposition 2.1]) [Project result].

Every SFT domain is a t -SFT domain.

Proof. Let D be SFT and let $P = P_t$ be a nonzero prime t -ideal. Since every ideal of D is SFT, there are a finitely generated $B \subseteq P$ and $e \geq 1$ such that $x^e \in B$ for every $x \in P$. But $P_t = P$ and $B \subseteq B_v$, so the same pair satisfies

$$x^e \in B_v \quad (x \in P_t).$$

Thus every nonzero prime t -ideal is t -SFT, and the prime- t -ideal criterion proves the assertion. $\square$

The second implication is strict, even for a familiar integrally closed domain.

Proposition 4.5 [Project result].

For every field k , the infinite polynomial ring

$$U = k[Z_1, Z_2, \dots]$$

is a t -SFT domain but is not an SFT ring.

Proof. Every nonzero element of U lies in a polynomial ring in finitely many variables, and its factorization there remains a factorization after more variables are adjoined. Thus U is a UFD. We verify the t -SFT assertion directly. Let I be a nonzero ideal and choose $0 \neq b \in I$. Only finitely many irreducibles $\pi_1, \dots, \pi_r$ divide b . For each i , choose $b_i \in I$ on which the nonnegative integer ord_{π_i} assumes its minimum over I , and put

$$F = (b, b_1, \dots, b_r) \subseteq I.$$

In a UFD, divisorial closure is read coefficientwise from the prime-divisor valuations:

$$H_v = \{x \in U : \text{ord}_{\pi}(x) \geq \inf_{0 \neq h \in H} \text{ord}_{\pi}(h) \text{ for every irreducible } \pi\}.$$

The displayed choice realizes every nonzero minimum occurring for I ; no irreducible outside the finite list can have a positive minimum because it does not divide b . Hence $F_v = I_v$. Since $F_v \subseteq I_t \subseteq I_v$, one has $I_t = F_v$. Therefore $(I, F, 1)$ is t -SFT data.

On the other hand, let $\mathfrak{n} = (Z_1, Z_2, \dots)$. Given a finitely generated $J \subseteq \mathfrak{n}$ and $e \geq 1$, all generators of J involve only finitely many variables. Choose Z_m outside that finite set and map all the variables occurring in the generators to zero. This homomorphism kills J but not Z_m^e , so $Z_m^e \notin J$. Thus $\mathfrak{n}$ is not SFT and neither is U . $\square$

We also need the descent statement for the t -operation. Unlike ordinary SFT descent, it is not a quotient argument.

Proposition 4.6 (Polynomial and formal t -SFT descent; using [11, Proposition 2.1]) [Project result].

Let D be a domain with quotient field K , and let $A = D[Z]$ or $A = D[[Z]]$. If A is a t -SFT domain, then D is a t -SFT domain.

Proof. We first record the contraction calculation that replaces quotient permanence. For every nonzero fractional ideal F of D ,

$$(FA)_v \cap K = F_v. \tag{4.7}$$

For $A = D[Z]$, coefficient comparison gives $(FA)^{-1} = F^{-1}A$ and then $(FA)_v = F_v A$. For $A = D[[Z]]$, Dobbs and Houston [11, Proposition 2.1] prove the stronger identities

$$(FD[[Z]])^{-1} = F^{-1}[[Z]], \quad (FD[[Z]])_v = F_v[[Z]],$$

which again give Equation (4.7) by taking constant series.

It follows that for every nonzero ideal I of D ,

$$(IA)_t \cap K = I_t. \quad (4.8)$$

Indeed, if $x \in I_t$, choose a nonzero finitely generated $F \subseteq I$ with $x \in F_v$. The stronger identities just used put the constant x in $(FA)_v \subseteq (IA)_t$. Conversely, if $x \in (IA)_t \cap K$, choose a nonzero finitely generated $J \subseteq IA$ with $x \in J_v$. Each generator of J is a finite A -linear combination of elements of I , so some nonzero finitely generated $C \subseteq I$ satisfies $J \subseteq CA$. Monotonicity and Equation (4.7) give

$$x \in J_v \cap K \subseteq (CA)_v \cap K = C_v \subseteq I_t,$$

proving Equation (4.8).

Now assume A is t -SFT and fix a nonzero ideal I of D . Apply the definition in A to IA : there are a nonzero finitely generated $J \subseteq IA$ and $e \geq 1$ such that

$$y^e \in J_v \quad (y \in (IA)_t).$$

As above, collect the finitely many coefficients from I needed to write the generators of J and obtain a nonzero finitely generated $C \subseteq I$ with $J \subseteq CA$. If $x \in I_t$, then Equation (4.8) places x in $(IA)_t$. Hence

$$x^e \in J_v \cap K \subseteq (CA)_v \cap K = C_v.$$

Thus (I, C, e) is t -SFT data. Since I was arbitrary, D is t -SFT. $\square$

Corollary 4.7 [Project result].

Let D be a domain and $A = D[Z]$ or $D[[Z]]$. For each of the properties SFT, t -SFT, and VSFT, the property for A implies the same property for D .

Proof. For SFT, pass to the quotient $A/(Z) \cong D$. The same quotient argument works for VSFT by lemma 4.8. The t -SFT assertion is Proposition 4.6. $\square$

Lemma 4.8 (Coykendall–Dutta [8, Theorem 4.6]). *VSFT descends to quotient rings.*

Proof. Let $\pi : A \rightarrow \bar{A}$ be a surjective homomorphism and assume that A is VSFT. Given an ideal $\bar{I}$ of $\bar{A}$, put $I = \pi^{-1}(\bar{I})$. Choose a finitely generated ideal $B \subseteq I$ and $n \geq 1$ with $I^n \subseteq B$. Applying π gives

$$\bar{I}^n = \pi(I^n) = \pi(I^n) \subseteq \pi(B) \subseteq \bar{I},$$

and $\pi(B)$ is finitely generated. Thus $\bar{I}$ is VSFT. $\square$

4.2. Failure of polynomial t -SFT ascent.

Theorem 4.9 [Project result].

For every prime p , the rings R_p and R_0 are t -SFT, but $R_p[X]$ and $R_0[X]$ are not t -SFT. More precisely, the upper to zero used in each construction is not a t -SFT ideal.

Proof. The coefficient rings R_p and R_0 are SFT domains, so Lemma 4.4 shows that they are t -SFT.

We give the stronger assertion about their polynomial rings. Let $D \in \{R_p, R_0\}$, put $L = \text{Frac}(D)$ and $P = D[X]$, and set

$$Q = \ker(P \rightarrow L, X \mapsto u).$$

This is the nonzero upper to zero from the construction. Suppose, toward a contradiction, that it has t -SFT data (Q, J, e) , with $J = (h_1, \dots, h_s) \subseteq Q$. Necessarily $J \neq 0$: otherwise the condition

applied to a nonzero element of $Q \subseteq Q_t$ would give a nonzero power equal to zero. Discarding zero generators, put

$$d = \max_i \deg h_i.$$

Because $Q \cap D = (0)$, every nonzero h_i has positive degree, so $d \geq 1$. Factor in $L[X]$ as

$$h_i = (X - u)\tilde{h}_i, \quad H = (\tilde{h}_1, \dots, \tilde{h}_s)P,$$

where H is a fractional ideal of P . If $h_i = \sum_j a_{ij}X^j$, synthetic division gives

$$\tilde{h}_i = \sum_{j \geq 1} a_{ij} \sum_{r=0}^{j-1} X^{j-1-r} u^r. \quad (4.9)$$

For every term on the right, $r \leq d-1$ and $u^r \in V_r \subseteq V_d$, whence $c^d u^r \in D$. It follows that $c^d H \subseteq P$. With inverse and v -closure now taken over P , this says $c^d \in H^{-1}$ and therefore

$$H_v = (H^{-1})^{-1} \subseteq c^{-d}P. \quad (4.10)$$

Since multiplication by the nonzero principal fractional ideal $(X - u)$ commutes with v -closure,

$$J_v = ((X - u)H)_v = (X - u)H_v. \quad (4.11)$$

First take $D = R_p$. For every $n \geq 0$, the polynomial

$$g_n = c(X - u)^{p^n} = cX^{p^n} - cu^{p^n}$$

belongs to Q , because $u^{p^n} \in V_1$. Choose $q = p^r \geq e$. The t -SFT assumption gives $g_n^e \in J_v$; since J_v is a P -ideal, it follows that $g_n^q \in J_v$. Using Equation (4.11), cancel one factor $X - u$ in $L[X]$ and then use Equation (4.10) to obtain

$$c^{q+d}(X - u)^{p^{n+r}-1} \in P. \quad (4.12)$$

In characteristic p ,

$$(X - u)^{p^N-1} = \frac{X^{p^N} - u^{p^N}}{X - u} = \sum_{j=0}^{p^N-1} u^j X^{p^N-1-j}.$$

Consequently Equation (4.12) gives $c^{q+d}u^j \in R_p$ for every $0 \leq j < p^{n+r}$. The coefficient test Equation (4.3) forces

$$u^j \in V_{q+d} \quad (0 \leq j < p^{n+r}).$$

Letting n tend to infinity gives $\mathbb{F}_p[u] \subseteq V_{q+d}$, contrary to Equation (4.1).

Now take $D = R_0$. For $m = 2^n$, the polynomial

$$g_n = c(X^m - u^m)$$

belongs to Q , because $u^{2^n} \in V_1$. Fix any integer $N \geq e$. As before, $g_n^N \in J_v$. Cancelling one factor $X - u$ and applying Equation (4.10) gives

$$c^{N+d} \frac{(X^m - u^m)^N}{X - u} \in P. \quad (4.13)$$

For $0 \leq j < mN$, put $s_j = \lfloor j/m \rfloor + 1$. Direct division gives

$$[X^j] \frac{(X^m - u^m)^N}{X - u} = (-1)^{N-s_j} \binom{N-1}{s_j-1} u^{mN-1-j}. \quad (4.14)$$

The binomial coefficient is nonzero in the characteristic-zero coefficient field. Applying Equation (4.3) coefficient by coefficient in Equation (4.13), and using that each nonzero scalar from the coefficient field is a unit, yields

$$1, u, \dots, u^{mN-1} \in V_{N+d}.$$

As $m = 2^n$ is unbounded, this contradicts $k[u] \not\subseteq V_{N+d}$. Thus Q is not t -SFT in either case, and neither polynomial ring is t -SFT. $\square$

Remark 4.10. There is no conflict with theorem 4.3. Indeed, the theorem just proved, together with the fact that R_p and R_0 are t -SFT, also shows that these direct counterexample domains cannot be integrally closed. Thus normality is a genuine boundary for the known positive theorem.

4.3. VSFT behavior of the characteristic-zero construction. The next proof uses the three-factor filtration estimate directly. An elementwise statement such as $z^3 \in cR_0$ would not by itself be enough to prove VSFT, because VSFT requires containment of an entire ideal power. The weaker conclusion that R_0 is VSFT also follows from the characteristic-zero polarization theorem of Coykendall–Dutta [8, Theorem 5.1]; the displayed cubic containment is the project strengthening proved here.

Proposition 4.11 [Project result].

The ring R_0 is VSFT. More precisely,

$$M_0^3 \subseteq cR_0. \quad (4.15)$$

Proof. Before localization, write

$$A_0 = \bigoplus_{n \geq 0} c^n V_n, \quad \mathfrak{m}_0 = \bigoplus_{n \geq 1} c^n V_n.$$

It is enough to check products of homogeneous elements. If $i, j, \ell \geq 1$ and $x \in V_i, y \in V_j, z \in V_\ell$, then

$$(c^i x)(c^j y)(c^\ell z) = c c^{i+j+\ell-1} xyz.$$

By Equation (4.5), $xyz \in V_{i+j+\ell-1}$, so the expression after the first factor c belongs to A_0 . Taking finite sums proves $\mathfrak{m}_0^3 \subseteq cA_0$, and localization proves Equation (4.15).

Let I be a nonzero proper ideal of R_0 and choose $0 \neq a \in I$. Then $I \subseteq M_0$, and the power-cofinality property of the construction provides $D \geq 1$ such that $c^D \in aR_0$. Therefore

$$I^{3D} = (I^3)^D \subseteq (M_0^3)^D \subseteq c^D R_0 \subseteq aR_0.$$

Thus $(I, (a), 3D)$ is VSFT data. The zero ideal and the unit ideal have the immediate witnesses $(0, 0, 1)$ and $(R_0, R_0, 1)$, respectively. Hence every ideal of R_0 is VSFT. $\square$

Corollary 4.12 [Project result].

The ring $R_0[X]$ is not VSFT. Thus VSFT does not ascend to a polynomial ring, even from a one-dimensional local domain containing $\mathbb{Q}$.

Proof. The construction proves that $R_0[X]$ is not SFT, whereas Equation (4.6) says that every VSFT ring is SFT. $\square$

We first record a useful equivalence for the characteristic-zero formal-series ring. The direct theorem following it determines both sides of the equivalence.

Proposition 4.13 (Coykendall–Dutta [8, Theorem 5.1]). *For a formal indeterminate Y ,*

$$R_0[[Y]] \text{ is SFT} \iff R_0[[Y]] \text{ is VSFT}.$$

The equivalence itself does not assert that either condition holds.

Proof. The implication from VSFT to SFT is Equation (4.6). Conversely, $R_0[[Y]]$ is a domain containing $\mathbb{Q}$. Coykendall and Dutta prove that every SFT ideal in such a domain is VSFT. If $R_0[[Y]]$ is an SFT ring, applying their theorem to each of its ideals makes it a VSFT ring. $\square$

4.4. Direct failure after one formal variable.

Theorem 4.14 (Formal t -SFT counterexamples) [Project result].

Let D be either R_p or R_0 , and put $B = D[[Y]]$. Then B is not a t -SFT domain. Consequently B is neither SFT nor VSFT.

More precisely, put

$$N_n = \begin{cases} p^n, & D = R_p, \\ 2^n, & D = R_0, \end{cases} \quad F_n = \frac{Y^{N_n} - u^{N_n}}{Y - u} = \sum_{r=0}^{N_n-1} Y^{N_n-1-r} u^r,$$

and

$$g_n = c(Y^{N_n} - u^{N_n}) = c(Y - u)F_n \in B.$$

Then the ideal $I = (g_n : n \geq 0)B$ is not a t -SFT ideal.

Proof. The two constructions have the common properties

$$u^r \in V_r, \quad c^r V_r \subseteq D, \quad k[u] \not\subseteq V_E \quad (E \geq 0),$$

and $u^{N_n} \in V_1$. Hence $g_n \in B$ for every n ; in particular $g_0 = c(Y - u) = cY - cu$ belongs to B .

Suppose that I had t -SFT data (I, J, e) . Write $J = (h_1, \dots, h_s)$. Necessarily $J \neq 0$, since a nonzero $g_n \in I \subseteq I_t$ cannot have a nonzero power in the zero ideal. Every h_i is a finite B -linear combination of the g_n , so there is N such that

$$J \subseteq I_N = (g_0, \dots, g_N)B.$$

In the quotient field of B , put

$$H = (h_1/g_0, \dots, h_s/g_0)B, \quad K_N = (F_0, \dots, F_N)B.$$

Thus $J = g_0 H$ and $H \subseteq K_N$. Set $d = N_N - 1$. If $j \leq N$, every coefficient of F_j is one of $1, u, \dots, u^d$, hence $c^d F_j \in B$. It follows that $c^d H \subseteq B$, or equivalently $c^d \in H^{-1}$. Therefore

$$H_v = (H^{-1})^{-1} \subseteq c^{-d} B, \quad J_v = (g_0 H)_v = g_0 H_v. \quad (4.16)$$

Here the second identity is the usual compatibility of v -closure with nonzero principal fractional scaling.

Every nonzero element $x \in I$ belongs to I_t , because $(x) = (x)_v \subseteq I_t$. Thus $g_m^e \in J_v$ for every m .

First let $D = R_p$. Choose $q = p^r \geq e$. Since J_v is a B -module, $g_m^q = g_m^{q-e} g_m^e \in J_v$. In characteristic p , $Y^{p^m} - u^{p^m} = (Y - u)^{p^m}$, so Equation (4.16) gives

$$c^{d+q-1} (Y - u)^{p^{m+r}-1} \in B. \quad (4.17)$$

The geometric identity

$$(Y - u)^{p^L-1} = \frac{Y^{p^L} - u^{p^L}}{Y - u} = \sum_{a=0}^{p^L-1} Y^{p^L-1-a} u^a$$

shows from Equation (4.17) that $c^{d+q-1} u^a \in R_p$ for $0 \leq a < p^{m+r}$. The coefficient test Equation (4.3) forces

$$u^a \in V_{d+q-1} \quad (0 \leq a < p^{m+r}).$$

Letting m grow gives $\mathbb{F}_p[u] \subseteq V_{d+q-1}$, contradicting the escape property.

Now let $D = R_0$ and write $M = 2^m$. From $g_m^e \in J_v$ and Equation (4.16) one obtains

$$c^{d+e-1} \frac{(Y^M - u^M)^e}{Y - u} \in B. \quad (4.18)$$

For $0 \leq a < Me$, put $s_a = \lfloor a/M \rfloor + 1$. Expanding $(Y^M - u^M)^{e-1} F_m$ gives

$$[Y^a] \frac{(Y^M - u^M)^e}{Y - u} = (-1)^{e-s_a} \binom{e-1}{s_a-1} u^{Me-1-a}. \quad (4.19)$$

The displayed binomial coefficient is a nonzero element of the characteristic-zero coefficient field k , hence a unit of R_0 . Applying Equation (4.3) to every coefficient in Equation (4.18) yields

$$1, u, \dots, u^{Me-1} \in V_{d+e-1}.$$

Since $M = 2^m$ is unbounded, this contradicts $k[u] \not\subseteq V_{d+e-1}$.

Thus I is not t -SFT in either characteristic. If B were SFT, then Lemma 4.4 would make it t -SFT. Finally, VSFT implies SFT by Equation (4.6). Both stronger properties therefore fail as asserted. $\square$

Remark 4.15. The proof does not substitute $Y = u$ in an arbitrary formal series. The telescoping factors $Y^{N_n} - u^{N_n}$ are polynomials already lying in $D[[Y]]$, and the contradiction comes entirely from their coefficients. This is why the direct formal argument is valid although the polynomial evaluation map used for the upper to zero has no formal-series extension.

4.5. Failure of VSFT in prime characteristic.

Proposition 4.16 [Project result].

For every prime p and every $N \geq 1$,

$$M_p^N \not\subseteq cR_p.$$

Consequently, R_p is not VSFT.

Proof. Fix $N \geq 1$. For $L \geq N$, put $F_L = \mathbb{F}_p(u^{p^L})$. The filtration estimate established in the construction, applied with $E = N - 1$, is

$$\dim_{F_L}(F_L V_{N-1}) \leq \sum_{r=0}^{N-1} ((N-1)L)^r = O_N(L^{N-1}). \quad (4.20)$$

For each N -element subset S of $\{0, \dots, L-1\}$, set

$$w_S = u^{\sum_{e \in S} p^e}.$$

The exponents in this family are distinct by uniqueness of base- p expansion, and they are all strictly smaller than p^L . Moreover, $1, u, \dots, u^{p^L-1}$ is an F_L -basis of $\mathbb{F}_p(u)$. Consequently the $\binom{L}{N}$ elements w_S are linearly independent over F_L . Since $\binom{L}{N}$ has degree N in L , while the bound in Equation (4.20) has degree at most $N-1$, for all sufficiently large L there is an S with

$$w_S \notin F_L V_{N-1}, \quad \text{and hence} \quad w_S \notin V_{N-1}.$$

For every $e \geq 0$, the element cu^{p^e} lies in M_p , since $u^{p^e} \in V_1$. Therefore

$$c^N w_S = \prod_{e \in S} cu^{p^e} \in M_p^N.$$

If this element belonged to cR_p , cancellation of c would give $c^{N-1}w_S \in R_p$. The coefficient test Equation (4.3) would then force $w_S \in V_{N-1}$, a contradiction. This proves $M_p^N \not\subseteq cR_p$ for every N .

Suppose now that M_p had VSFT data

$$M_p^e \subseteq B = (b_1, \dots, b_t) \subseteq M_p.$$

By Equation (4.4), $b_i^p \in cR_p$ for every i . Put $k = t(p-1) + 1$. Every monomial of total degree k in $b_1, \dots, b_t$ has some exponent at least p , so

$$B^k \subseteq (b_1^p, \dots, b_t^p) \subseteq cR_p.$$

It follows that $M_p^{ek} \subseteq B^k \subseteq cR_p$, contradicting the first part with $N = ek$. Thus the maximal ideal is not VSFT, and neither is R_p . $\square$

Corollary 4.17 [Project result].

For every prime p , neither $R_p[X]$ nor $R_p[[Y]]$ is VSFT.

Proof. The polynomial ring $R_p[X]$ is not SFT, so it cannot be VSFT by Equation (4.6). Evaluation at $Y = 0$ is a surjection $R_p[[Y]] \rightarrow R_p$. If $R_p[[Y]]$ were VSFT, then lemma 4.8 would make R_p VSFT, contrary to Proposition 4.16. $\square$

4.6. Logical specializations of broader variants. The following negative statements use only exact specializations of the definitions; they make no assertion for a fixed nontrivial multiplicative set or for a separately chosen extension of a semistar operation.

Proposition 4.18 [Project result].

The following universally quantified polynomial-ascent assertions are false.

- (i) *For every multiplicative set S , every S -SFT ring D has an S -SFT polynomial ring $D[X]$.*
- (ii) *For every multiplicative set S , every uniformly S -SFT ring D has a uniformly S -SFT polynomial ring $D[X]$.*
- (iii) *For every semistar operation, polynomial ascent holds in a manner which includes the identity operation d_D on D and the identity operation $d_{D[X]}$ on $D[X]$.*

Proof. For $S = \{1\}$, both the S -SFT and uniformly S -SFT definitions reduce exactly to ordinary SFT: the only possible multiplier from S is 1. Taking $D = R_0$, Proposition 4.11 in particular gives that D is SFT, whereas the construction gives that $D[X]$ is not SFT. This disproves (i) and (ii).

For the identity semistar operation d , the d -SFT definition is exactly ordinary SFT. The pair d_{R_0} and $d_{R_0[X]}$ therefore turns the assertion in (iii) into the already false ordinary SFT implication $R_0 \Rightarrow R_0[X]$. This proves (iii). Notice that the argument says nothing about a fixed nonidentity operation, and such a question must also specify which operation on $D[X]$ extends the one on D . $\square$

4.7. Summary of the three variants. The conclusions proved in this section are summarized below.

Ring	SFT	t -SFT	VSFT
R_p	yes	yes	no
$R_p[X]$	no	no	no
$R_p[[Y]]$	no	no	no
R_0	yes	yes	yes
$R_0[X]$	no	no	no
$R_0[[Y]]$	no	no	no

Remark 4.19. For reference, the elementary prime chains

$$(0) \subsetneq M_p[[Y]] \subsetneq M_p[[Y]] + (Y), \quad (0) \subsetneq M_0[[Y]] \subsetneq M_0[[Y]] + (Y),$$

Indeed, coefficientwise reduction identifies the quotients by $M_p[[Y]]$ and $M_0[[Y]]$ with formal-series rings over the corresponding domain quotients; hence the middle ideals in the displayed chains are prime. The final ideals are prime because their quotients are the residue fields. Thus they give the elementary bounds

$$\dim R_p[[Y]] \geq 2, \quad \dim R_0[[Y]] \geq 2.$$

The much stronger dimensions and cardinal prime-chain bounds proved later are recorded in Theorem 6.12. They are not consequences of confusing $D[X][[Y]]$ with $D[[Y]][X]$; the direct block construction is essential.

5. FORMAL POWER SERIES: STABILITY, DIMENSION, AND PRIME EXTENSION

For a ring A , write $A[[\mathbf{X}]] = A[[X_1, \dots, X_s]]$ when $\mathbf{X}$ is a finite tuple of formal variables. The phrase “ n -stably SFT” means that $A[[X_1, \dots, X_n]]$ is SFT. Quotienting by the last variable proves

$$n\text{-stably SFT} \implies (n-1)\text{-stably SFT}.$$

The reverse implication from zero to one variable is false [7]. The implication from one to two variables remains unresolved in this project; none of the reductions below is presented as a solution of that problem.

5.1. Arnold's baseline and horizontal primes. We begin with the two strong-convergence facts used throughout. Arnold's dimension theorem [2, Theorem 1] says that

$$A \text{ not SFT} \implies \dim A[[Y]] = \infty. \quad (5.1)$$

The later cardinal strengthening says that $A[[Y]]$ actually contains a prime chain of size at least $2^{\aleph_1}$ whenever A is non-SFT; the precise published statement is recorded in theorem 6.14. For an SFT ring A and $P \in \text{Spec } A$, the formal radical identity and its finite-variable strong-convergence form [2, 5] are

$$\sqrt{PA[[\mathbf{X}]]} = P[[\mathbf{X}]], \quad P[[\mathbf{X}]] \subseteq Q \text{ if } Q \cap A = P. \quad (5.2)$$

Moreover $P[[\mathbf{X}]]$ is an SFT ideal. These cited results are the formal-series input behind the deductions that follow. Notice also that $A[[Y]]$ SFT always implies A SFT, by quotienting by (Y) . Thus a non-SFT coefficient ring gives both a non-SFT and, by Equation (5.1), infinite-dimensional formal-series ring.

Theorem 5.1 (Horizontal-prime criterion) [Project result].

Let A be SFT. Then $A[[Y]]$ is SFT if and only if, for every $P \in \text{Spec } A$, every prime

$$\mathfrak{q} \subseteq (A/P)[[Y]], \quad \mathfrak{q} \cap (A/P) = 0,$$

is an SFT ideal.

Proof. Necessity follows by passing first to the quotient by $P[[Y]]$ and then to the chosen prime ideal. Conversely, take $Q \in \text{Spec } A[[Y]]$ and put $P = Q \cap A$. By Equation (5.2), the ideal $P[[Y]]$ is SFT and is contained in Q . The image

$$\overline{Q} = Q/P[[Y]] \subseteq (A/P)[[Y]]$$

has zero contraction and is SFT by hypothesis. The lifting lemma Lemma 2.5 lifts its data across the SFT ideal $P[[Y]]$, so Q is SFT. Arnold's prime criterion theorem 2.2 now makes $A[[Y]]$ SFT. $\square$

Thus a formal obstruction is horizontal in a residue-domain series ring; it is not merely a coefficient ideal extended vertically.

Corollary 5.2 (Reduction to domains) [Project result].

The universal implication

$$R[[X]] \text{ SFT} \implies R[[X, Y]] \text{ SFT}$$

holds for all rings if and only if it holds for domains.

Proof. Only one direction needs proof. Assume the implication for domains and suppose $R[[X]]$ is SFT. Let $Q \in \text{Spec } R[[X, Y]]$ and choose a minimal prime $p \subseteq Q \cap R$. The quotient

$$(R/p)[[X]] \cong R[[X]]/p[[X]]$$

is an SFT domain, so the domain hypothesis makes $(R/p)[[X, Y]]$ SFT. The ideal $p[[X, Y]]$ is SFT by Equation (5.2); lifting the SFT data of $Q/p[[X, Y]]$ proves that Q is SFT. The prime criterion finishes the proof. $\square$

5.2. Valuation conductors and the normal boundary. For a finite-dimensional SFT domain R , let $g_s(R)$ denote the supremum of the dimensions of all s -variable residue-domain generic formal fibres, as in Equation (5.5) below.

Theorem 5.3 (Valuation-conductor recursion) [Project result].

Let R be a normal finite-dimensional SFT domain and let $V \neq R$ be a proper valuation overring in $\text{Frac } R$. Suppose that

$$0 \neq P = (R : V) = \{z \in \text{Frac } R : zV \subseteq R\}.$$

Then P is a nonzero prime of both R and V , the domain R/P is normal, and for every $s \geq 1$,

$$g_s(R) = \max\{s, g_s(R/P)\}. \quad (5.3)$$

Consequently

$$\dim R[[X_1, \dots, X_s]] < \infty \iff \dim(R/P)[[X_1, \dots, X_s]] < \infty.$$

Proof. We first prove that P is radical in V . If $z \in V$ and $z^n \in P$, then for every $y \in V$,

$$(zy)^n = z^n y^n \in R.$$

Thus zy is integral over the normal domain R , so $zy \in R$. Hence $zV \subseteq R$ and $z \in P$. A proper radical ideal of a valuation domain is prime, and contraction makes P prime in R .

Every prime $\mathfrak{q}$ of R is comparable with P . If $P \not\subseteq \mathfrak{q}$, choose $c \in P \setminus \mathfrak{q}$. Since $cV \subseteq R$, one has $R_c = V_c$. The prime $\mathfrak{q}_c$ therefore corresponds to a valuation prime $\mathfrak{h}$ avoiding c . Valuation primes are linearly ordered, and $c \in P \setminus \mathfrak{h}$ forces $\mathfrak{h} \subsetneq P$. Because $\mathfrak{h} \subseteq P \subseteq R$, contraction gives $\mathfrak{q} = \mathfrak{h} \subsetneq P$.

Put $W = V_P$. Valuation primes are divided, so $PW = P$, and every prime of W below P is literally a prime of R . Extending an R -SFT witness for each such prime to W and using Arnold's prime criterion shows that W is an SFT valuation domain. Its prime chains are chains in R , so it is finite-dimensional.

Fix $\mathfrak{q} \subsetneq P$ and choose $0 \neq c \in P \setminus \mathfrak{q}$. Coefficientwise,

$$c(W/\mathfrak{q}) \subseteq P/\mathfrak{q} \subseteq R/\mathfrak{q}.$$

Multiplication by this one c clears every coefficient of every formal series over $W/\mathfrak{q}$. After all nonzero constants are inverted, this gives an isomorphism

$$((R/\mathfrak{q})[[\mathbf{X}]])_{(R/\mathfrak{q}) \setminus \{0\}} \cong ((W/\mathfrak{q})[[\mathbf{X}]])_{(W/\mathfrak{q}) \setminus \{0\}}. \quad (5.4)$$

The positive-dimensional finite-dimensional SFT valuation domain $W/\mathfrak{q}$ has a nonidempotent height-one prime. To see the nonidempotence directly, a finite ideal in a valuation domain is principal, say bW . If an SFT prime H with witness (H, bW, n) were idempotent, then total divisibility of principal ideals would turn the elementwise containments $x^n \in bW$ into $H^n \subseteq bW$. Hence $H = H^n = bW$, impossible for a nonzero proper idempotent principal ideal in a domain. The valuation-localization theorem of Kang and Park [13] therefore identifies the ring on the right as an s -dimensional Noetherian UFD. Thus every residue fibre strictly below P has dimension exactly s . The primes containing P correspond to $\text{Spec}(R/P)$ and give precisely the fibres defining $g_s(R/P)$. Since $(0) \subsetneq P$, at least one lower fibre occurs, proving Equation (5.3).

It remains to prove normality of the quotient. Lift an element of $\text{Frac}(R/P)$ integral over R/P to $w \in V_P$, and lift a monic equation to $f(T) \in R[T]$. Then $f(w) \in PV_P = P$, while $f(T) - f(w) \in R[T]$ is monic and kills w . Normality of R gives $w \in R$, so its residue belongs to R/P . Finally apply Theorem 5.13 to R and R/P . $\square$

Corollary 5.4 (Consequences of the recursion) [Project result].

Suppose there exists a normal finite-dimensional SFT domain R with

$$\dim R[[X]] = \infty,$$

and choose one of minimal possible dimension. Then R satisfies

$$(R : V) = 0$$

for every proper valuation overring $V \neq \text{Frac } R$. In particular it is nonconductive. More generally, any finite iteration of Equation (5.3) ending in a field proves finite formal dimension.

Proof. A nonzero conductor would produce a smaller-dimensional normal SFT quotient R/P with the same finiteness failure, contradicting minimality. The characterization of conductive domains by nonzero conductors into proper valuation overrings [10] gives the second statement. $\square$

The same argument identifies exactly why valuation-ideal pullbacks do not produce a normal counterexample.

Proposition 5.5 (Normal valuation-ideal pullbacks) [Project result].

Let V be a valuation domain with maximal ideal $\mathfrak{n}$, let $0 \neq I \subsetneq V$ be an ideal, and let $k \subseteq V$ be a field with $k \cap \mathfrak{n} = 0$. Put $R = k + I$. Then R is normal if and only if $I = P$ is prime in V and k is relatively algebraically closed in

$$\kappa(P) = \text{Frac}(V/P).$$

In that case $R = k + P$ is a PVD with associated valuation domain $W = V_P$, and

$$R \text{ is SFT} \iff W \text{ is SFT}.$$

Consequently, a finite-dimensional SFT ring in this class belongs to the PVD/APVD stratum. Rings in this stratum are known to be stable under formal-series extension [16].

Proof. Choose $0 \neq a \in I$. Since every $v \in V$ equals $(av)/a$, the rings R and V have the same quotient field. Suppose R is normal. If $z \in \sqrt{I}$, then a power of z lies in R , so normality gives $z \in R$; because $z \in \mathfrak{n}$, its k -component is zero and $z \in I$. Thus I is radical, hence prime in the valuation domain. Put $P = I$. If $\lambda \in \kappa(P)$ is algebraic over k , lift it to $w \in V_P$. A monic $f \in k[T]$ annihilating its residue has $f(w) \in P$; the monic polynomial $f(T) - f(w) \in R[T]$ kills w . Normality gives $w \in R$ and hence $\lambda \in k$.

Conversely, if $z \in \text{Frac } R$ is integral over R , it is integral over the normal valuation domain V , hence belongs to V . Its residue modulo P is algebraic over k , hence lies in k ; so $z \in k + P = R$.

For the SFT equivalence, witnesses extend from R to W . Conversely, if $Q \subseteq P$ and $z^n = \sum_i f_i w_i$ is a W -witness, then

$$z^{n+1} = \sum_i f_i(z w_i)$$

is an R -witness because $z w_i \in Q \subseteq P \subseteq R$. Apply the prime criterion. Formal stability and the dimension formulas now follow from the published SFT-PVD/APVD theorem [16]. $\square$

Proposition 5.6 (A rank-two normal PVD) [Project result].

Let k be algebraically closed, $L = k(t)$, and $K = L((x))(y)$. Let V be the valuation domain for

$$v(f) = (\text{ord}_y(f), \text{ord}_x(\text{lc}_y(f))) \in \mathbb{Z} \text{ lex } \mathbb{Z},$$

and put $R = k + xV$. Then R is a two-dimensional normal, non-Noetherian, non-Prüfer SFT PVD, yet for every $s \geq 1$,

$$R[[X_1, \dots, X_s]] \text{ is SFT}, \quad \dim R[[X_1, \dots, X_s]] = 3s.$$

Proof. The maximal ideal of V is xV , its residue field is L , and k is relatively algebraically closed in L ; normality follows from Proposition 5.5. The prime chain is

$$(0) \subsetneq \mathfrak{p} = \{z : \text{the first coordinate of } v(z) \text{ is positive}\} \subsetneq xV.$$

For $z \in \mathfrak{p}$ and $w \in xV$, valuation comparison gives

$$z^2 \in yR, \quad w^2 \in xR.$$

These are SFT witnesses for the two nonzero primes. The quotient $xV/x^2V \cong k(t)$ is infinite-dimensional over $R/xV \cong k$, so R is not Noetherian. Also $t, t^{-1} \notin R$, so the local domain is not a valuation domain and hence not Prüfer. The residue-field branch of the PVD dimension theorem [16] gives 3s and SFT stability. $\square$

We next give the normal one-dimensional stress test that survives the conductor reduction.

Theorem 5.7 (A nonconductive normal LPVD stress test) [Project result].

Let

$$K = k(X_i, Y_i : i \geq 1), \quad F_i = k(X_j, Y_j : j \neq i), \quad E_i = F_i(Y_i), \\ V_i = E_i[X_i]_{(X_i)}, \quad R_i = F_i + X_i V_i, \quad D = \bigcap_{i \geq 1} R_i \subseteq K, \quad M_i = D \cap X_i V_i.$$

Then:

- (i) D is a one-dimensional normal SFT domain of finite character,
 $\text{Frac } D = K, \quad D_{M_i} = R_i, \quad \text{Spec } D = \{(0)\} \cup \{M_i : i \geq 1\},$
and $\sqrt{X_i D} = M_i, \quad M_i^2 \subseteq X_i D.$
- (ii) D is non-Noetherian, non-Prüfer, non-PvMD, and nonconductive; indeed $(D : V_i) = 0$ for every i .
- (iii) If $T = \bigcap_i V_i$, then $T = D^{\text{cic}}$ is a PID, $D \subseteq T$ is unibranched, and $(D : T) = 0$. The domain D is not a GPVD and is not completely integrally closed.
- (iv) For $G_D = (D \setminus \{0\})^{-1} D[[Z]]$,

$$G_D = D[[Z]][X_i^{-1} : i \geq 1].$$

Equivalently, $\sum_n a_n Z^n \in G_D$ precisely when there are a finite $H \subset \mathbb{N}$ and integers c_i such that

$$a_n \in R_i \quad (i \notin H), \quad v_i(a_n) \geq c_i \quad (i \in H)$$

for every n .

- (v) Each local generic fibre $G_i = (R_i \setminus \{0\})^{-1} R_i[[Z]] = V_i[[Z]][X_i^{-1}]$ is a one-dimensional Noetherian UFD, and the generic fibre G_T has dimension one, but

$$G_D \subsetneq G_T \subseteq \bigcap_i G_i, \quad \Phi(Z) = \sum_{n \geq 1} Y_n Z^n \in G_T \setminus G_D.$$

The dimension of G_D is not determined in this project.

- (vi) For arbitrary primes $\mathfrak{p}_i \in \text{Spec } G_i$ and a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the assembled prime

$$P_{\mathcal{U}} = \{f \in G_D : \{i : f \in \mathfrak{p}_i\} \in \mathcal{U}\}$$

is always (0) or (Z) .

Thus D is a stress test, not a counterexample.

Proof. Each V_i is a DVR with residue field E_i , and F_i is relatively algebraically closed in the purely transcendental extension $E_i = F_i(Y_i)$. Thus Proposition 5.5 makes every R_i a normal one-dimensional PVD. Every nonzero rational function involves only finitely many pairs (X_i, Y_i) and is a unit of all remaining R_i . This local finiteness makes the intersection normal and gives $\text{Frac } D = K$ because $X_i, X_i Y_i \in D$.

To compute D_{M_i} , take $z \in R_i$. It fails to belong to only finitely many $R_j, j \neq i$. Multiplying by sufficiently large powers of those X_j clears the failures; their product is a unit in R_i . Hence $z \in D_{M_i}$. Conversely every element of $D \setminus M_i$ is a unit of R_i , proving $D_{M_i} = R_i$.

For $i \neq j$, $X_i + X_j$ is a unit in every R_ℓ , hence in D . If $z, w \in M_i$, then zw/X_i belongs to R_i and to every other R_j because X_i is a unit there. Thus $M_i^2 \subseteq X_i D$ and $\sqrt{X_i D} = M_i$. A nonzero $a \in D$ is a nonunit at only finitely many indices. A monomial in sufficiently high powers of their

X_i lies in aD . Every nonzero prime therefore contains some X_i , hence M_i , and cannot contain a second M_j because $X_i + X_j$ is a unit. This proves the spectrum formula, finite character, and the displayed SFT witnesses.

The maximal ideal of R_i is not finitely generated because $X_i V_i / X_i^2 V_i \cong E_i$ is infinite-dimensional over F_i ; thus D is not Noetherian. Since $Y_i, Y_i^{-1} \notin R_i$, it is not Prüfer. For $0 \neq c \in D$, choose $j \neq i$ outside the finite variable support of c . Then $Y_j \in V_i$ but $cY_j \notin R_j$, proving $(D : V_i) = 0$ and nonconduciveness. A one-dimensional SFT PvMD is Prüfer, so D is not a PvMD.

Put $N_i = T \cap X_i V_i$. The same clearing argument gives $T_{N_i} = V_i$ and $N_i = X_i T$. The monomial argument shows that these are all nonzero primes of T . Hence every prime is finitely generated; Cohen's criterion makes T Noetherian, and the one-dimensional normal domain with principal maximal ideals is a PID. If z is almost integral over D , valuation inequalities force $z \in V_i$ for every i , hence $z \in T$. Conversely, for $z \in T$, multiplication by the product of the finitely many X_i at which $z \notin R_i$ puts every z^n in D . Thus $T = D^{\text{cic}}$. The elements $Y_i \in T \setminus D$ show that D is not completely integrally closed, and the preceding conductor argument also gives $(D : T) = 0$. The unique contractions $N_i \mapsto M_i$ show unbranchedness. Any associated Prüfer overring for a GPVD would localize to every V_i and hence equal T ; the required nonzero common ideal would contradict $(D : T) = 0$. Thus D is not a GPVD.

Let S_0 be the monoid generated by the X_i . The monomial divisibility argument says that for each $0 \neq d \in D$ some $m \in S_0$ lies in dD ; therefore $1/d = (m/d)/m$ and $G_D = D[[Z]][S_0^{-1}]$. Reading valuations coefficientwise proves the restricted-denominator criterion in (iv), and conversely multiplication by $\prod_{i \in H} X_i^{N_i+1}$ clears any prescribed uniform lower bounds.

Multiplication by X_i clears every coefficient from V_i into R_i , which gives $G_i = V_i[[Z]][X_i^{-1}]$ and dimension one. Since T is a one-dimensional SFT Prüfer domain, its generic fibre also has dimension one. For fixed i , every coefficient Y_n with $n \neq i$ lies in F_i , and multiplication by X_i clears Y_i , so $\Phi \in G_i$ for all i and $\Phi \in G_T$. It violates the finite-exception condition in (iv), hence $\Phi \notin G_D$. This strict inclusion is exactly why the one-dimensional local and complete-integral-closure fibres do not determine $\dim G_D$.

Finally, for $0 \neq f = \sum a_n Z^n \in G_D$, let $r = \text{ord}_Z(f)$. Outside finitely many indices, all coefficients belong to R_i and the leading coefficient a_r belongs to $F_i^\times$. Hence $f = Z^r u_i$ with $u_i \in R_i[[Z]]^\times$ for all but finitely many i . Modulo a nonprincipal ultrafilter, membership of f in $\mathfrak{p}_i$ is therefore equivalent either to $f = 0$, or to $r > 0$ and $Z \in \mathfrak{p}_i$. This gives exactly (0) or (Z) . The displayed set is prime because it is the kernel of the diagonal map to the ultraproduct of the domains $G_i/\mathfrak{p}_i$. $\square$

Corollary 5.8 (The remaining one-dimensional normal target) [Project result].

Let A be a one-dimensional normal SFT domain. Then A has finite character, and all finite-variable formal-dimension questions reduce to the single zero generic fibre

$$(A \setminus \{0\})^{-1} A[[X_1, \dots, X_s]].$$

If A is a PvMD or is completely integrally closed, then

$$\dim A[[X_1, \dots, X_s]] = s + 1.$$

Hence a one-dimensional normal counterexample not excluded by this project must be nonconductive, non-PvMD, and not completely integrally closed. The domain in Theorem 5.7 realizes these necessary features, but its zero-fibre dimension is not determined here.

Proof. An SFT ring has Noetherian spectrum. For $0 \neq a \in A$, the closed zero-dimensional subspace $V(a)$ is Noetherian and therefore has only finitely many points; this is finite character. Every nonzero residue domain is a field, whose s -variable generic fibre has dimension s , so Theorem 5.13 leaves only the zero fibre.

In a one-dimensional PvMD every maximal ideal is a t -prime, because it is minimal over a nonzero principal ideal. The PvMD localization criterion therefore makes every maximal localization a

valuation domain, so A is Prüfer. The SFT-Prüfer formula [15] gives $s + 1$. A completely integrally closed SFT domain is Krull [20]; in dimension one it is Dedekind and the same formula applies. Nonconduciveness follows from Corollary 5.4. $\square$

5.3. Two positive mechanisms.

Definition 5.9. A domain D with quotient field K has the *countable common-denominator property* if, for every sequence $(a_i)_{i \geq 0}$ in K , there is $0 \neq d \in D$ such that $da_i \in D$ for every i .

Proposition 5.10 (Common denominators) [Project result].

If D has the countable common-denominator property, then

$$(D \setminus \{0\})^{-1} D[[Y]] = K[[Y]],$$

and the only primes of $D[[Y]]$ contracting to zero are (0) and (Y) . Consequently, if A is SFT and every residue domain A/P has this property, then $A[[Y]]$ is SFT.

Proof. For $F = \sum a_i Y^i \in K[[Y]]$, a common denominator d gives $dF \in D[[Y]]$, proving the displayed equality. A zero-contraction prime localizes to a prime of the DVR $K[[Y]]$, hence to (0) or (Y) . Prime localization correspondence shows that contraction returns the original prime, and both resulting primes are principal. Apply Theorem 5.1 residue domain by residue domain. $\square$

Finite jets do not see the obstruction: if A is SFT, then $A[[Y]]/(Y^m)$ is SFT by the nilpotent-thickening lemma. The witnesses may depend on m , and therefore do not supply one exponent for an infinite series. Likewise, a prime of $R[[X, Y]]$ containing a formal coordinate is controlled by quotienting and lifting; a bad prime must avoid every coordinate obtainable by a formal automorphism.

Theorem 5.11 (Bounded-radicial stability) [Project result].

Let $A \subseteq B$ have characteristic $p > 0$, and suppose that one integer $e \geq 0$ satisfies $b^{p^e} \in A$ for every $b \in B$. For finite tuples of formal variables $\mathbf{Y}$ and polynomial variables $\mathbf{X}$, put

$$C = A[[\mathbf{Y}]][\mathbf{X}], \quad E = B[[\mathbf{Y}]][\mathbf{X}].$$

If C is SFT, then E is SFT.

Proof. Frobenius acts coefficientwise and kills all mixed terms, so $F^{p^e} \in C$ for every $F \in E$. Hence $C \subseteq E$ is uniformly radicial, and Lemma 2.8 applies. $\square$

Corollary 5.12 (Normal algebraic positive class) [Project result].

Let A be a normal Noetherian domain of characteristic $p > 0$, let $K = \text{Frac } A$, and let L/K be algebraic. Let F be the relative separable closure of K in L . Assume that $[F : K] < \infty$, the normalization C of A in F is module-finite over A , and L/F is purely inseparable of bounded exponent e . If B is the normalization of A in L , then

$$B[[\mathbf{Y}]][\mathbf{X}] \text{ is SFT,} \quad \dim B[[\mathbf{Y}]][\mathbf{X}] = \dim A + |\mathbf{Y}| + |\mathbf{X}|.$$

Proof. The finite A -algebra C is normal and Noetherian. For $b \in B$, bounded inseparability gives $b^{p^e} \in F$. This element is integral over C , hence belongs to C by normality. Apply Theorem 5.11 to $C \subseteq B$. The extension of the mixed formal-polynomial rings is integral, while the source is Noetherian; lying over, going up, and incomparability give the dimension formula from the usual Noetherian formula for C and the equality $\dim C = \dim A$. $\square$

5.4. Generic formal fibres and exact dimension bounds. Let A be a finite-dimensional SFT ring, $d = \dim A$, and $s \geq 1$. For $P \in \operatorname{Spec} A$ define

$$G_P^{(s)} = ((A/P)[[X_1, \dots, X_s]])_{(A/P) \setminus \{0\}}, \quad g_P^{(s)} = \dim G_P^{(s)}, \quad g_s = \sup_P g_P^{(s)}. \quad (5.5)$$

Theorem 5.13 (Exact generic-fibre reduction) [Project result].

With the preceding notation,

$$g_s \leq \dim A[[X_1, \dots, X_s]], \quad (5.6)$$

$$\dim A[[X_1, \dots, X_s]] \leq \sup_{P_0 \subsetneq \dots \subsetneq P_r} \left(r + \sum_{j=0}^r g_{P_j}^{(s)} \right) \leq d + (d+1)g_s. \quad (5.7)$$

Consequently

$$\dim A[[X_1, \dots, X_s]] < \infty \iff g_s < \infty.$$

The right side requires one uniform bound over all residue domains.

Proof. Take a strict finite chain $Q_0 \subsetneq \dots \subsetneq Q_N$ in $A[[\mathbf{X}]]$. Its distinct contractions form

$$P_0 \subsetneq \dots \subsetneq P_r, \quad r \leq d,$$

and equal contractions occur in $r+1$ contiguous blocks. In the block over P_j , Equation (5.2) permits quotienting by $P_j[[\mathbf{X}]]$. Localizing at the nonzero elements of A/P_j sends that block to a strict chain in $G_{P_j}^{(s)}$; strictness is preserved by the order isomorphism on primes disjoint from the multiplicative set. The block contributes at most $g_{P_j}^{(s)}$ inclusions. Adding the r inclusions between blocks gives

$$N \leq r + \sum_{j=0}^r g_{P_j}^{(s)},$$

which proves the two upper bounds.

Conversely, a finite chain in $G_P^{(s)}$ contracts through localization and lifts through $A[[\mathbf{X}]] \rightarrow (A/P)[[\mathbf{X}]]$, giving a chain of the same length in $A[[\mathbf{X}]]$. This proves the lower bound and the equivalence. $\square$

Theorem 5.14 (Finite-extension transfer of generic fibres) [Project result].

Let $A \subseteq B$ be a module-finite inclusion of domains and put

$$G_A = (A \setminus \{0\})^{-1} A[[\mathbf{X}]], \quad G_B = (B \setminus \{0\})^{-1} B[[\mathbf{X}]].$$

Then G_B is module-finite and integral over G_A , and

$$\dim G_B = \dim G_A.$$

Also, for every fixed finite tuple $\mathbf{X}$,

$$A[[\mathbf{X}]] \text{ is SFT} \iff B[[\mathbf{X}]] \text{ is SFT}, \quad \dim A[[\mathbf{X}]] = \dim B[[\mathbf{X}]].$$

Proof. Fixed A -module generators of B generate $B[[\mathbf{X}]]$ coefficientwise over $A[[\mathbf{X}]]$. The final SFT equivalence and dimension equality are therefore exactly the finite-module results of Theorem 2.11 and Corollary 2.12; no finite-invariance proof is repeated here.

For the generic fibres, set $S = A \setminus \{0\}$. The ring $S^{-1}B[[\mathbf{X}]]$ is finite over G_A . The constant ring $S^{-1}B$ is a finite-dimensional domain over $\operatorname{Frac} A$, hence is a field. Thus every nonzero element of B is already a unit after localizing at S , and

$$S^{-1}B[[\mathbf{X}]] = (B \setminus \{0\})^{-1} B[[\mathbf{X}]] = G_B.$$

The extension is finite and integral. Lying over and going up give one dimension inequality, and incomparability gives the other. $\square$

Corollary 5.15 (Residual N-1 normalization) [Project result].

Let D be a finite-dimensional residually N-1 SFT domain. For $P \in \operatorname{Spec} D$, let B_P be the normalization of D/P in its quotient field. Then

$$\dim((D/P)[[\mathbf{X}]]_{(D/P) \setminus \{0\}}) = \dim(B_P[[\mathbf{X}]]_{B_P \setminus \{0\}}),$$

and

$$\dim D[[\mathbf{X}]] < \infty \iff \sup_{P \in \operatorname{Spec} D} \dim(B_P[[\mathbf{X}]]_{B_P \setminus \{0\}}) < \infty.$$

Proof. Residual N-1 says precisely that B_P is finite over D/P . Apply Theorem 5.14 to every residue domain and then Theorem 5.13. Finite normalization removes nonnormality from each individual fibre; it does not create the required uniform bound as P varies. $\square$

5.5. Weierstrass and axis-trace control.**Theorem 5.16 (The Weierstrass trap)** [Project result].

Assume $R[[T]]$ is SFT, put $B = R[[X, Y]]$, and let $Q \in \operatorname{Spec} B$. Suppose that after a continuous R -algebra coordinate change $(X, Y) \mapsto (U, V)$, the prime Q contains a series F satisfying

$$F(U, 0) = U^d u(U), \quad d \geq 1, \quad u(U) \in R[[U]]^\times.$$

Then Q is an SFT ideal.

Proof. Multiplying by the unit $u(U)^{-1}$ reduces to $F \equiv U^d \pmod{V}$. Put $C = R[[V]]$, so $B = C[[U]]$. Weierstrass division is valid here without a Noetherian hypothesis. Modulo V , divide uniquely by U^d . If quotient and remainder have been constructed modulo V^N , divide the error modulo V^{N+1} by U^d ; the congruence $F \equiv U^d \pmod{V}$ gives the unique correction. Coefficientwise completeness and separatedness give

$$C[[U]] \cong \varprojlim_N C[[U]]/V^N C[[U]], \quad \bigcap_N V^N C[[U]] = 0.$$

Passing to the inverse limit shows that every class modulo F has a unique representative of U -degree less than d . Hence

$$B/(F) \cong C^d$$

as a C -module. The ring $C \cong R[[T]]$ is SFT, so module-finite ascent Theorem 2.10 makes $B/(F)$ SFT. The ideal $Q/(F)$ is then SFT, and Lemma 2.5 lifts its data across the principal ideal (F) . $\square$

A bad prime is therefore doubly non-Weierstrass after every formal coordinate change. The condition cannot be weakened to “monic in one variable”: for example, $\mathbb{Z}[[Y]]/(Y-2) \cong \mathbb{Z}_2$ is not module-finite over $\mathbb{Z}$. The topological nilpotence in Theorem 5.16 is essential.

Theorem 5.17 (Axis trace and topological Nakayama) [Project result].

Let D be a domain and let $\mathfrak{q} \in \operatorname{Spec} D[[Y]]$ have zero contraction to D .

- (i) If $Y \in \mathfrak{q}$, then $\mathfrak{q} = (Y)$.
- (ii) If $Y \notin \mathfrak{q}$, put

$$C_{\mathfrak{q}} = \{F(0) : F \in \mathfrak{q}\} \subseteq D.$$

Then evaluation induces

$$\mathfrak{q}/Y\mathfrak{q} \cong C_{\mathfrak{q}},$$

and

$$\mathfrak{q} \text{ is finitely generated} \iff C_{\mathfrak{q}} \text{ is finitely generated.}$$

Proof. If $Y \in \mathfrak{q}$, quotienting by (Y) leaves the zero prime of D . Now assume $Y \notin \mathfrak{q}$. The kernel of evaluation on $\mathfrak{q}$ is

$$\mathfrak{q} \cap YD[[Y]] = Y\mathfrak{q} :$$

if $YH \in \mathfrak{q}$, primality and $Y \notin \mathfrak{q}$ give $H \in \mathfrak{q}$. This proves the module isomorphism and one finite-generation implication.

Conversely, suppose $C_{\mathfrak{q}} = (c_1, \dots, c_r)$ and choose $f_i \in \mathfrak{q}$ with $f_i(0) = c_i$. Given $f \in \mathfrak{q}$, put $r_0 = f$. Recursively choose $d_{im} \in D$ and $r_{m+1} \in \mathfrak{q}$ so that

$$r_m(0) = \sum_i d_{im}c_i, \quad r_{m+1} = \frac{r_m - \sum_i d_{im}f_i}{Y}.$$

For every N ,

$$f = \sum_i \left(\sum_{m=0}^{N-1} d_{im}Y^m \right) f_i + Y^N r_N.$$

With $H_i = \sum_{m \geq 0} d_{im}Y^m$, coefficientwise passage to the limit gives $f = \sum_i H_i f_i$. Thus the f_i generate $\mathfrak{q}$. $\square$

In particular, a residually Noetherian SFT ring A has $A[[Y]]$ SFT: apply Theorems 5.1 and 5.17. Conversely, a bad horizontal prime over an SFT domain has a nonzero, non-finitely-generated SFT trace. The trace cannot be zero, because repeated division by Y and separatedness would force the prime itself to be zero.

Corollary 5.18 (Trace-chain rigidity) [Project result].

If $\mathfrak{q} \subsetneq \mathfrak{q}'$ are horizontal primes of $D[[Y]]$ which avoid Y and have the same trace, then that common trace is not finitely generated.

Proof. If the trace were generated by $c_1, \dots, c_r$, choose lifts $f_i \in \mathfrak{q}$. Apply the recursive proof of Theorem 5.17 to an arbitrary $f \in \mathfrak{q}'$, using these same lifts. The coefficientwise limit expresses f in $(f_1, \dots, f_r) \subseteq \mathfrak{q}$, forcing $\mathfrak{q}' \subseteq \mathfrak{q}$, a contradiction. $\square$

Proposition 5.19 (Structure of horizontal traces) [Project result].

Let $\mathfrak{q} \subseteq D[[Y]]$ be horizontal and avoid Y , and set $C = C_{\mathfrak{q}}$. Then:

- (i) *if $ab \in C$ and $aD + bD = D$, then $a \in C$ or $b \in C$; the same holds for $\sqrt{C}$, and $\text{Spec}(D/C)$ is connected;*
- (ii) *for $T = D[[Y]]/\mathfrak{q}$ and $y = Y + \mathfrak{q}$,*

$$C = D \cap yT;$$

- (iii) *if D is complete for the x -adic topology and $C_N = \{F(x^N) : F \in \mathfrak{q}\}$, then*

$$C_N = D \cap (y - x^N)T.$$

Proof. For (i), take $F \in \mathfrak{q}$ with $F(0) = ab$. Recursively factor $F = GH$ with $G(0) = a$ and $H(0) = b$: after lower coefficients have been chosen, the next equation is $ah_m + bg_m = r_m$, solvable because $aD + bD = D$. Primality puts G or H in $\mathfrak{q}$. If $ab \in \sqrt{C}$, apply the same argument to $a^N b^N \in C$; comaximality is preserved by powers. Finally an idempotent $\bar{e} \in D/C$ satisfies $e(1 - e) \in C$ and $eD + (1 - e)D = D$, so $e \in C$ or $1 - e \in C$.

For (ii), if $F \in \mathfrak{q}$, write $F = F(0) + YG$; reduction in T puts $-F(0)$ in yT . Conversely, if $d = yt$ with $d \in D$, lift t to $G \in D[[Y]]$; then $d - YG \in \mathfrak{q}$ has constant term d .

For (iii), x -adic convergence gives formal division

$$F(Y) - F(x^N) = (Y - x^N)G_F(Y).$$

This proves one inclusion after reduction in T . Conversely, lift an expression $d = (y - x^N)t$ to $d - (Y - x^N)G \in \mathfrak{q}$ and evaluate at x^N . $\square$

5.6. Trace recursion and its exact obstructions.

Theorem 5.20 (Trace-power recursion and trichotomy) [Project result].

Let D be a domain, $A = D[[Y]]$, and let $\mathfrak{q} \in \text{Spec } A$ have zero contraction and avoid Y . Suppose that its trace $C = C_{\mathfrak{q}}$ has SFT data (C, B, n) , where $B = (b_1, \dots, b_t)$. Choose $g_i \in \mathfrak{q}$ with $g_i(0) = b_i$ and put $J = (g_1, \dots, g_t)$. For every $f \in \mathfrak{q}$ and $r \geq 1$ there is $f_r \in \mathfrak{q}$ such that

$$f^{n^r} \equiv Y^{1+n+\dots+n^{r-1}} f_r \pmod{J}. \quad (5.8)$$

Put $S = D \setminus \{0\}$ and

$$H = \sqrt{JA_S} \cap A.$$

Then $\sqrt{J} \subseteq H \subseteq \mathfrak{q}$. If $\mathfrak{q}$ is not SFT, exactly one of the following occurs:

- (1) horizontal branching: $H \subsetneq \mathfrak{q}$;
- (2) denominator saturation: $\sqrt{J} \subsetneq H = \mathfrak{q}$;
- (3) unbounded nilpotence: $\sqrt{J} = H = \mathfrak{q}$, but no one exponent sends all of $\mathfrak{q}$ into J .

Proof. Put $f_0 = f$. Since $f_r(0) \in C$, choose $a_{ir} \in D$ with

$$f_r(0)^n = \sum_i a_{ir} b_i.$$

The element $f_r^n - \sum_i a_{ir} g_i$ belongs to $\mathfrak{q}$ and has zero constant term, so it equals $Y f_{r+1}$. Since $Y \notin \mathfrak{q}$, primality gives $f_{r+1} \in \mathfrak{q}$. Raising the congruence $f_r^n \equiv Y f_{r+1} \pmod{J}$ successively to the n th power proves Equation (5.8).

Localization gives $\sqrt{J} \subseteq H \subseteq \mathfrak{q}$. If $H \subsetneq \mathfrak{q}$, choose $f \in \mathfrak{q} \setminus H$. Prime existence in A_S gives a prime over JA_S avoiding $f/1$; its contraction is another horizontal prime. It cannot contain Y , because the first recursion would then put f in it. This is horizontal branching.

If $H = \mathfrak{q}$ but $\sqrt{J} \neq \mathfrak{q}$, every $f \in \mathfrak{q}$ satisfies $s_f f^{m_f} \in J$ for some $s_f \in S$, while some f has no power in J ; this is precisely denominator saturation. If $\sqrt{J} = \mathfrak{q}$, then $\mathfrak{q}/J$ is nil. A uniform nilpotence exponent would be SFT data $(\mathfrak{q}, J, N)$, so badness is exactly the failure of such an exponent. The three possibilities exhaust the two inclusions. $\square$

Proposition 5.21 (Torsion and minimal components) [Project result].

In the notation of Theorem 5.20,

$$H/\sqrt{J} = T_D(A/\sqrt{J}),$$

the coefficient-torsion submodule. If $\text{Min}_h(J)$ denotes the minimal primes over J disjoint from S and $\text{Min}_v(J)$ the remaining minimal primes, then

$$H = \bigcap_{\mathfrak{p} \in \text{Min}_h(J)} \mathfrak{p}, \quad \sqrt{J} = H \cap \bigcap_{\mathfrak{p} \in \text{Min}_v(J)} \mathfrak{p}.$$

Thus denominator saturation is removal of vertical components by localization, whereas horizontal branching means either a nonminimal horizontal prime or at least two incomparable horizontal minimal components.

Proof. If $f \in H$, then $s f^m \in J$ for some $s \in S$, so $(s f)^m \in J$ and $s f \in \sqrt{J}$. Conversely, $s f \in \sqrt{J}$ implies $s^r f^r \in J$ for some r , hence $f/1 \in \sqrt{JA_S}$. This proves the torsion formula. Minimal primes over JA_S correspond exactly to minimal primes over J disjoint from S . Intersecting them and contracting proves the first component formula; the second is the usual intersection of all minimal primes over J . $\square$

5.7. Moving axes and low-dimensional residue domains. Assume in this subsection that $A_0 = R[[X]]$ is SFT, take $P \in \text{Spec } A_0$, and put

$$D = A_0/P, \quad x = X + P.$$

For $\Gamma \in \text{Spec } R[[X]]$, write

$$\epsilon(\Gamma) = \{F(0) : F \in \Gamma\} \subseteq R.$$

Theorem 5.22 (Moving-axis obstruction) [Project result].

If $D[[Y]]$ has a non-SFT zero-contraction prime, then $x \neq 0$ and there is $\Gamma \in \text{Spec } R[[X]]$ such that

$$P \subsetneq \Gamma, \quad X \notin \Gamma, \quad \epsilon(P) \subsetneq \epsilon(\Gamma).$$

Consequently

$$X \in P \quad \text{or} \quad \epsilon(P) \text{ maximal in } R \implies D[[Y]] \text{ is SFT.}$$

Proof. The radical strong-convergence theorem makes P closed in the X -adic topology. Hence D is complete and separated for the x -adic topology. If $x = 0$, put $p = P \cap R$. Then $P = p[[X]] + (X)$ and $D \cong R/p$; after renaming the variable, $D[[Y]]$ is the SFT quotient $R[[X]]/p[[X]]$, a contradiction. Thus $x \neq 0$.

Let $\mathfrak{q} \subseteq D[[Y]]$ be a bad zero-contraction prime. For $N \geq 1$, completeness defines an automorphism

$$\tau_N(F)(Y) = F(Y + x^N),$$

whose inverse substitutes $Y - x^N$. Every coefficient is an x -adically convergent sum. Put

$$\mathfrak{q}_N = \tau_N(\mathfrak{q}), \quad C_N = \{F(x^N) : F \in \mathfrak{q}\}.$$

The prime $\mathfrak{q}_N$ remains bad and horizontal. It avoids Y , for otherwise Theorem 5.17 would give $\mathfrak{q}_N = (Y)$. Its trace C_N is therefore a nonzero, non-finitely-generated SFT ideal. Evaluation gives

$$C_N + x^N D = C_{\mathfrak{q}} + x^N D. \quad (5.9)$$

We claim $x \notin \sqrt{C_N}$. Otherwise choose $G \in \mathfrak{q}_N$ with $G(0) = x^r$. Lift through $R[[X, Y]] \twoheadrightarrow D[[Y]]$ to $\tilde{G}$ in the inverse image of $\mathfrak{q}_N$. If $h(X) = \tilde{G}(X, 0)$, then $h - X^r \in P$, so $h - X^r$, viewed as constant in Y , belongs to $P[[Y]]$ and to that inverse image. Subtracting it produces an element whose restriction to $Y = 0$ is X^r . The Weierstrass trap makes the inverse-image prime SFT and hence makes $\mathfrak{q}_N$ SFT, a contradiction. The same conclusion holds for $C_{\mathfrak{q}}$: if $x^r \in C_{\mathfrak{q}}$, choose $N > r$ and use Equation (5.9) to write

$$c_N = x^r(1 - x^{N-r}d) \in C_N.$$

The parenthesis is a unit because $x \in \text{Jac } D$.

Choose a prime $M_N \supseteq C_N$ avoiding x , and let Γ be its inverse image in A_0 . Then $P \subsetneq \Gamma$ and $X \notin \Gamma$. Plain inclusion gives $\epsilon(P) \subseteq \epsilon(\Gamma)$. Equality is impossible by the equal-trace-chain theorem [5]: in an SFT formal-series ring, a strict prime extension with equal axis trace must contain the variable. This proves the strict trace increase.

If $\epsilon(P)$ is maximal, apply the horizontal-prime criterion to every prime factor above P . A factor strictly above P either has a strictly larger proper trace, impossible by maximality, or has equal trace and hence contains X . In the latter case its residue domain is a quotient of R , whose one-variable formal ring is an SFT quotient of $R[[X]]$. The case $X \in P$ was handled at the start. Thus no bad horizontal prime remains. $\square$

Theorem 5.23 (One-dimensional residue-domain theorem) [Project result].

In the preceding setting, if $x = 0$ or $D[1/x]$ is a field, then $D[[Y]]$ is SFT. In particular,

$$\dim D \leq 1 \implies D[[Y]] \text{ is SFT.}$$

Hence a bad residue-domain reduction for one-to-two-variable stability has dimension at least two.

Proof. The case $x = 0$ is in Theorem 5.22. Suppose $x \neq 0$ and $D[1/x]$ is a field. A bad horizontal prime would produce a nonzero trace C_N with $x \notin \sqrt{C_N}$. A prime over C_N avoiding x would be a nonzero prime of D surviving in $D[1/x]$, a contradiction.

To apply Theorem 5.1 to every residue domain, take $M \in \text{Spec } D$. The case $M = 0$ was just handled. If $M \neq 0$, the field hypothesis forces $x \in M$. Its inverse image Γ in $R[[X]]$ contains X , and

$$D/M \cong R/(\Gamma \cap R).$$

Thus $(D/M)[[Y]]$ is, after renaming the variable, an SFT quotient of $R[[X]]$. The horizontal criterion finishes the proof.

Finally $x \in \text{Jac } D$. If D has dimension at most one, every nonzero prime is maximal and contains x , so $D[1/x]$ has only the zero prime and is a field. $\square$

Corollary 5.24 [Project result].

If $R[[X]]$ is SFT and $\dim R[[X]] \leq 1$, then $R[[X, Y]]$ is SFT.

Proof. Every prime factor of $R[[X]]$ has dimension at most one, so apply Theorem 5.23 and then Theorem 5.1. $\square$

5.8. Moving divisors, trapping, and finite incidence. Retain a bad prime $\mathfrak{q} \subseteq D[[Y]]$ from the moving-axis setting, and put

$$T = D[[Y]]/\mathfrak{q}, \quad y = Y + \mathfrak{q}, \quad B = D[1/x], \quad E = T[1/x], \quad I_N = C_N B.$$

Proposition 5.25 (Moving principal divisors) [Project result].

For every $N \geq 1$,

$$I_N = B \cap (y - x^N)E.$$

The proper principal ideals $(y - x^N)E$ are pairwise comaximal. Therefore a bad $\mathfrak{q}$ forces $\text{Max } E$ to be infinite.

Proof. Formal division gives $F(Y) - F(x^N) = (Y - x^N)G_F(Y)$, proving one contraction inclusion. Conversely, if

$$\frac{d}{x^r} = (y - x^N) \frac{h}{x^s}$$

in E , lift h to $H \in D[[Y]]$. Injectivity of $T \rightarrow T[1/x]$ gives

$$x^s d - x^r (Y - x^N) H \in \mathfrak{q}.$$

Evaluation at x^N puts $x^s d$ in C_N , proving the reverse inclusion. The contraction I_N is proper because $x \notin \sqrt{C_N}$. If $M > N$, the difference of the generators is

$$x^N (x^{M-N} - 1),$$

a unit of B because x is inverted and $1 - x^{M-N}$ is a unit in D . Thus the principal ideals are pairwise comaximal, and maximal ideals chosen above them are distinct. $\square$

Theorem 5.26 (Trapping and finite incidence) [Project result].

The following are equivalent:

- (i) *there is no nonzero prime $M \subset D$, avoiding x , with $\mathfrak{q} \subseteq M[[Y]]$;*
- (ii) *for every unbounded $\Lambda \subseteq \mathbb{N}$, $\sum_{N \in \Lambda} I_N = B$;*
- (iii) *every $\mathfrak{n} \in \text{Max } B$ contains only finitely many I_N .*

If these conditions hold, then for every L there are a finite $F_L \subseteq \{N : N \geq L\}$ and $s_L \geq L$ such that

$$x^{s_L} \in \sum_{N \in F_L} C_N.$$

If B is semilocal, the conditions fail: every bad prime is trapped in some nonzero $M[[Y]]$ with $x \notin M$.

Proof. If a tail sum is proper, put it in a maximal ideal $\mathfrak{n}$ of B and contract to $M \subset D$. It is nonzero and avoids x . For $F = \sum_i a_i Y^i \in \mathfrak{q}$, suppose a_j is the first coefficient outside M . Since $F(x^N) \in M$ for every N in the unbounded set, cancellation in the domain D/M gives

$$\bar{a}_j \in x^N(D/M)$$

for unbounded N . The residue-domain origin makes D/M x -adically separated, so $\bar{a}_j = 0$, a contradiction. Hence $\mathfrak{q} \subseteq M[[Y]]$. Conversely such a trap contains every C_N because M is x -adically closed, so every tail sum is proper. This proves (i) $\Leftrightarrow$ (ii); a subset of positive integers is infinite exactly when it is unbounded, giving (iii).

Under (ii), 1 lies in the localized sum of every tail, so finitely many terms and a common power of x give the displayed containment. Moreover

$$\sum_{N \geq L} C_N + x^L D = C_{\mathfrak{q}} + x^L D$$

by Equation (5.9). If $s_L < L$, cancellation would put x^{s_L} times a unit in $C_{\mathfrak{q}}$, contrary to $x \notin \sqrt{C_{\mathfrak{q}}}$.

If B is semilocal, choose for each N a maximal ideal containing I_N . One maximal occurs for an unbounded set of N , so the argument above gives a trap. $\square$

Proposition 5.27 (Two positive generic strata) [Project result].

In the moving-axis setting:

- (i) If $B = D[1/x]$ is semilocal and finite-dimensional and every nonzero prime of D avoiding x is invertible, then no bad horizontal prime exists.
- (ii) Suppose B is Noetherian and $H : x^\infty$ is finitely generated for every finitely generated $H \subseteq D$. If one moving trace C_N is x -saturated (in particular, if it is primary and avoids x), then the horizontal prime is not bad.

Proof. For (i), Theorem 5.26 traps a putative bad prime in $M[[Y]]$ for a nonzero invertible prime M avoiding x . Since $M[[Y]] = MD[[Y]]$, put $\mathcal{L} = (MD[[Y]])^{-1}\mathfrak{q} \subseteq D[[Y]]$. Then $\mathfrak{q} = (MD[[Y]])\mathcal{L}$. For $z \in \mathcal{L}$, choose $0 \neq m \in M$. One has $mz \in \mathfrak{q}$, while $m \notin \mathfrak{q}$ by zero contraction; primality gives $z \in \mathfrak{q}$. Hence $\mathcal{L} \subseteq \mathfrak{q}$ and

$$\mathfrak{q} = (MD[[Y]])\mathfrak{q}.$$

Iteration puts all coefficients of $\mathfrak{q}$ in $K = \bigcap_{r \geq 1} M^r$. If $K \neq 0$, localize at M , where M is principal. A prime over K avoiding its generator contracts to a nonzero prime $L \subsetneq M$ avoiding x . Repeating gives an infinite descending chain of nonzero primes in $D[1/x]$, contradicting finite dimension. Hence $K = 0$ and $\mathfrak{q} = 0$, impossible for a bad prime.

For (ii), $I_N = C_N B$ is finitely generated. Choose finite $H \subseteq C_N$ with $HB = I_N$. Saturation gives

$$C_N = I_N \cap D = HB \cap D = H : x^\infty,$$

which is finitely generated by hypothesis. This contradicts Theorem 5.17, since every moving trace of a bad prime is non-finitely-generated. A primary ideal avoiding x is automatically x -saturated. $\square$

Corollary 5.28 (Residual generic-stratum criterion) [Project result].

Let $D = R[[X]]/P$ and $x = X + P$. Suppose that, for every prime quotient D/L with $x \notin L$, the generic stratum $(D/L)[1/x]$ is semilocal and finite-dimensional and every nonzero prime of D/L avoiding x is invertible as an ideal of D/L . Then $D[[Y]]$ is SFT. If this condition holds for every prime P of $R[[X]]$, then $R[[X, Y]]$ is SFT.

Proof. If $x \in L$, the residue-domain series ring is an SFT quotient of $R[[X]]$, as in Theorem 5.23. If $x \notin L$, apply Proposition 5.27 to D/L . The horizontal-prime criterion, first for D and then for $R[[X]]$, finishes the proof. $\square$

5.9. Reciprocal denominators and the trace spectrum. The anti-Archimedean terminology and its power-series background come from Anderson–Kang–Park [1]; the exact equivalence below is the project result proved here.

Theorem 5.29 (Reciprocal-denominator criterion) [Project result].

Let D be a domain, $K = \text{Frac } D$, $A = D[[Y]]$, $S = D \setminus \{0\}$, and $L = S^{-1}A \subseteq K[[Y]]$. The following are equivalent:

- (i) every $F \in D[[Y]]$ with $F(0) \neq 0$ has $F^{-1} \in L$;
- (ii) $\text{Spec } L = \{(0), (Y)L\}$;
- (iii) the only primes of $D[[Y]]$ contracting to zero are (0) and (Y) ;
- (iv) D is anti-Archimedean:

$$\bigcap_{n \geq 1} b^n D \neq 0 \quad (0 \neq b \in D).$$

Under these conditions L is a DVR with uniformizer Y .

Proof. Assume (i) and let $0 \neq Q \in \text{Spec } L$. Choose an element of Q of least Y -order and write it $Y^r G$ with $G(0) \neq 0$. If $Y \notin Q$, then $G \in Q$. Clear one denominator so that $sG \in D[[Y]]$. Its constant is nonzero, so (i) makes sG , and hence G , a unit of L , a contradiction. Therefore $Y \in Q$, and $L/(Y)L \cong K$ gives $Q = (Y)L$. Conversely, under (ii), a series of nonzero constant cannot be a nonunit, because every nonunit belongs to the unique maximal ideal $(Y)L$. This proves (i) $\Leftrightarrow$ (ii). Localization correspondence and $(Y)L \cap D[[Y]] = (Y)$ prove (ii) $\Leftrightarrow$ (iii). Localization correspondence and $(Y)L \cap D[[Y]] = (Y)$ prove (ii) $\Leftrightarrow$ (iii).

Write

$$F = b + \sum_{i \geq 1} a_i Y^i, \quad F^{-1} = \sum_{n \geq 0} h_n Y^n, \quad b \neq 0.$$

Coefficient recursion gives $b^{n+1} h_n \in D$. If $0 \neq d \in \bigcap_j b^j D$, then $dh_n \in D$ for every n , so $dF^{-1} \in D[[Y]]$ and $F^{-1} \in L$. Thus (iv) implies (i). Conversely, apply (i) to

$$(b - Y)^{-1} = \sum_{n \geq 0} b^{-n-1} Y^n.$$

A common nonzero denominator belongs to every $b^n D$. Finally, under (i), every nonzero element of L is a power of Y times a unit; choosing the least order in an ideal shows that every nonzero ideal is a power of (Y) . $\square$

The common-denominator property in Proposition 5.10 is stronger: it identifies L with all of $K[[Y]]$. The reciprocal theorem needs only enough denominators for inverses and still determines the complete horizontal spectrum.

Theorem 5.30 (Exact horizontal-trace spectrum) [Project result].

Put

$$\mathcal{A}(D) = \left\{ 0 \neq b \in D : \bigcap_{n \geq 1} b^n D = 0 \right\}.$$

Then

$$\mathcal{A}(D) = \bigcup_{\substack{q \in \text{Spec } D[[Y]] \\ q \cap D = 0, Y \notin q}} (C_q \setminus \{0\}).$$

For every horizontal trace,

$$\sqrt{C_q} \setminus \{0\} \subseteq \mathcal{A}(D).$$

Proof. If $0 \neq b \in C_{\mathfrak{q}}$, choose $F \in \mathfrak{q}$ with $F(0) = b$. A nonzero element in every $b^n D$ would make F a unit of L by the proof of Theorem 5.29, contradicting the properness of $S^{-1}\mathfrak{q}$. Thus $b \in \mathcal{A}(D)$.

Conversely, if $b \in \mathcal{A}(D)$, the displayed geometric-series expansion shows that $b - Y$ is a nonunit of L . Put it in a maximal ideal Q . The ideal Q cannot contain Y , because then it would contain the unit $b = (b - Y) + Y$. Contracting Q gives a horizontal prime whose trace contains b . If a nonzero power of a lies in a trace and $\bigcap a^n D$ were nonzero, the same would hold for that power; this proves the radical refinement. $\square$

Corollary 5.31 (Residual bounded-element criteria) [Project result].

Assume $R[[X]]$ is SFT. For each

$$P \in \operatorname{Spec} R[[X]] \quad \text{with} \quad X \notin P,$$

put $D = R[[X]]/P$ and $x = X + P$.

- (i) If every nonzero non-finitely-generated SFT ideal $I \subseteq D$ with $x \notin \sqrt{I}$ contains a nonzero b satisfying $\bigcap_n b^n D \neq 0$, then $R[[X, Y]]$ is SFT.
- (ii) Let

$$\mathcal{U}(D) = \{0 \neq b : \bigcap_n b^n D \neq 0\}.$$

Suppose every nonzero prime of D avoiding x meets $\mathcal{U}(D)$ and every nonzero non-finitely-generated SFT ideal avoiding x in its radical is x -saturated. Then again $R[[X, Y]]$ is SFT.

Proof. A failure produces, by Theorems 5.1 and 5.22, a bad horizontal prime over some such D . Its trace C is nonzero, non-finitely-generated, SFT, and satisfies $x \notin \sqrt{C}$. In (i), the hypothesis produces $b \in C \setminus \mathcal{A}(D)$, contradicting Theorem 5.30.

For (ii), note first that $\mathcal{U}(D)$ is multiplicatively closed. The prime hypothesis says that

$$D[1/x, \mathcal{U}(D)^{-1}] \text{ is a field.}$$

Extending C to that field gives

$$x^r u \in C$$

for some $r \geq 0$ and $u \in \mathcal{U}(D)$. The trace-spectrum theorem gives $C \cap \mathcal{U}(D) = \emptyset$, so $r \geq 1$. Hence $u \in C : x^\infty$ but $u \notin C$: every surviving trace is strictly non- x -saturated. This contradicts the second hypothesis. $\square$

The same argument gives a sharper reduced statement. For a radical ideal J , one has $J : x^m = J : x$ for all $m \geq 1$. Thus, under the generic-prime hypothesis in part (ii), every bad trace satisfies

$$\sqrt{C} : x \supsetneq \sqrt{C}.$$

In $D/\sqrt{C}$ there are nonzero classes of x and of a bounded element u whose product is zero. If $x \notin \sqrt{C}$, distinct incomparable minimal primes respectively contain u and x . This coefficient-side branching is compatible with the connectedness in Proposition 5.19; it is not the horizontal branching in Theorem 5.20.

5.10. Boundary models for the trace mechanisms. The following examples delimit the preceding arguments. None is a counterexample to one-to-two-variable SFT stability.

Proposition 5.32 (Sharp boundary models) [Project result].

The three trace-recursion mechanisms and the finite-incidence alternatives can occur in otherwise very well behaved coefficient settings:

- (i) Let $D = k[[x]]$, $A = D[[Y]]$, and $\mathfrak{q} = (Y - x)$. Then $C_{\mathfrak{q}} = (x)$. With $J = x\mathfrak{q}$ one has $\sqrt{J} \subsetneq \sqrt{JA_{D \setminus \{0\}}} \cap A = \mathfrak{q}$; with $J = ((Y - x)(Y - x^2))$, horizontal localization has two incomparable minimal components.

- (ii) *A rank-one nondiscrete valuation domain with value group $\mathbb{Q}$ has a principal ideal J whose radical is its maximal ideal, but the nilpotence exponents in $\sqrt{J}/J$ are unbounded.*
- (iii) *For $D = k[[x, z]]$ and $\mathfrak{q} = (Y - z)$, the moving traces are $C_N = (x^N - z)$. After inverting x , the coefficient ring is a one-dimensional PID and the C_N become pairwise distinct maximal ideals; two axes already cover a power of x . The prime $\mathfrak{q}$ is principal and good.*
- (iv) *There is a finite-character Bézout domain with infinitely many pairwise comaximal, primary, non-finitely-generated SFT ideals, each with a one-point maximal support. This coefficient-side model does not realize the ideals as the affine family $D \cap (y - x^N)T$ in one common quotient.*
- (v) *Even the exact provenance $D = R[[X]]/P_0$ does not force a nonzero non-finitely-generated radical SFT ideal avoiding x to contain a bounded element, to be x -saturated, or to be primary. Such an ideal exists in an SFT domain D for which $D[[Y]]$ is nevertheless SFT.*

Proof. For (i), substitution $Y \mapsto x$ has kernel $\mathfrak{q}$. The trace is (x) . For $J = x(Y - x)A$, localization makes x a unit and hence has radical $\mathfrak{q}$, while $Y - x \notin \sqrt{J}$ by reduction modulo x . For $J = (Y - x)(Y - x^2)A$, localization has the incomparable prime components $(Y - x)$ and $(Y - x^2)$. These are exactly denominator saturation and the minimal-component branch, even though D is a complete Noetherian DVR and $\mathfrak{q}$ is principal.

For (ii), choose x of value one and let $J = xV$. Its radical is the maximal ideal M . Given N , choose y with $0 < v(y) < 1/N$; then $y \in M$ but $y^N \notin J$. The domain V is not SFT, so this is only a nilpotence boundary model.

For (iii), evaluation at x^N gives $C_N = (x^N - z)$. The localization $k[[x, z]][1/x]$ is a one-dimensional Noetherian UFD, hence a PID, and

$$k[[x, z]][1/x]/(x^N - z) \cong k((x)).$$

Thus every localized trace is maximal. For $M > N$,

$$(x^N - z) + (x^M - z) \ni x^N(1 - x^{M-N}),$$

so two axes cover a power. This verifies all assertions while $\mathfrak{q}$ remains principal.

For (iv), let $G = \bigoplus_{n \geq 1} \mathbb{Q}e_n$, let K be the fraction field of $k[G]$, let v_n be the n th Gauss valuation, and put

$$B = \bigcap_n \{a \in K : v_n(a) \geq 0\}, \quad I_n = \{b \in B : v_n(b) > 1\}.$$

Every element has nonzero value at only finitely many indices. Factoring out coordinatewise minima shows that every two-generated ideal is principal, so B is Bézout; its maximal ideals are $M_n = \{v_n > 0\}$, and B_{M_n} is the corresponding $\mathbb{Q}$ -valued valuation domain. Hence B has finite character. The ideal I_n is M_n -primary and is not finitely generated because its localization is an open-cut ideal. Yet, with a monomial $X_n^{3/2}$,

$$a^2 \in X_n^{3/2}B \quad (a \in I_n),$$

so it is SFT. Different I_n are comaximal. The construction contains no common element y giving the affine contractions required by moving traces.

For (v), take the rank-two valuation domain V with value group $\mathbb{Z} \text{lex } \mathbb{Z}$ and primes $(0) \subsetneq P \subsetneq M$. Choose y, z with values $(1, 0)$ and $(0, 1)$. Then P is non-finitely-generated but $a^2 \in yV$ for $a \in P$, while $M = zV$ is principal; hence V is SFT. It is Prüfer, so $D = V[[X]]$ and $D[[Y]]$ are SFT [15]. Set $I = P[[X]]$ and $J = XI$. Then

$$J = (X) \cap I, \quad J : X = I \supsetneq J,$$

so J is radical, non-finitely-generated, SFT, nonprimary, and not X -saturated, while $X \notin J$. Localizing V at P gives a DVR whose first valuation coordinate shows

$$\bigcap_{n \geq 1} F^n D = 0 \quad (0 \neq F \in I),$$

so no nonzero member of J is bounded. Nevertheless $D[[Y]]$ is SFT. Thus coefficient-ideal properties alone do not manufacture one bad horizontal prime. $\square$

For the explicit rings R_p and R_0 , the preceding boundary models are no longer the last word. Polynomial evaluation at $X = u$ still does not define a homomorphism on arbitrary formal series, but the separate telescoping-block proof in Theorem 4.14 proves the negative SFT and t -SFT results for the two direct rings. The cardinal block construction in Theorem 6.12 proves infinite ordinary dimension and much more. These later arguments do not alter the role of the boundary models above; they use additional special filtration structure of the explicit rings.

5.11. Going up and incomparability. For a coefficient extension $R \subseteq T$, put

$$A_n = R[[X_1, \dots, X_n]], \quad B_n = T[[X_1, \dots, X_n]].$$

If T is finite as an R -module, the same module generators generate B_n over A_n coefficientwise. Thus the extension is finite and integral and has lying over, going up, and incomparability. This is the prime-theoretic part of Corollary 2.12; we do not repeat its finite-module proof.

Proposition 5.33 (Transcendental field obstruction) [Project result].

Let $k \subseteq K$ be fields and suppose $e \in K$ is transcendental over k . For every $n \geq 2$, the extension

$$k[[X_1, \dots, X_n]] \subseteq K[[X_1, \dots, X_n]]$$

has neither incomparability nor going up.

Proof. It is enough to use X_1, X_2 . Put $f = eX_1 + X_2$. The ideal (f) in the large formal ring is prime, because the coordinate change $X_2 \mapsto X_2 - eX_1$ carries it to (X_2) . We claim

$$(f)K[[X_1, \dots, X_n]] \cap k[[X_1, \dots, X_n]] = 0.$$

If a nonzero series over k were divisible by f , each of its homogeneous components would be divisible by f in the polynomial ring over K . Substituting $X_2 = -eX_1$ in a nonzero component would give a polynomial relation for e over $k[X_1, X_3, \dots, X_n]$, impossible. Thus the strict prime chain $(0) \subsetneq (f)$ has equal contractions, and incomparability fails.

For going up, start with $(0) \subsetneq (X_1)$ in the small ring and the prime (f) over zero. If a prime $Q \supseteq (f)$ contracted to (X_1) , then $X_1 \in Q$, hence $X_2 = f - eX_1 \in Q$, although $X_2 \notin (X_1)$ in the small ring. This is impossible. $\square$

Merely assuming that e is “not quadratic” is insufficient. If e is algebraic of degree $d > 2$ with minimal polynomial μ , then the nonzero homogenization

$$X_1^d \mu(-X_2/X_1) \in k[X_1, X_2]$$

is divisible by $eX_1 + X_2$. Thus the old linear-form example requires a transcendental element; finite algebraic extensions are instead covered by module-finite integrality.

Theorem 5.34 (Exact field classification) [Project result].

Let $k \subseteq K$ be fields and $n \geq 2$. In characteristic $p > 0$, put

$$E_r = k(K^{p^r}) \subseteq K.$$

For $A = k[[X_1, \dots, X_n]]$ and $B = K[[X_1, \dots, X_n]]$, the following are equivalent:

- (i) $A \subseteq B$ is integral;
- (ii) $A \subseteq B$ has going up;
- (iii) $A \subseteq B$ has incomparability;

(iv) $[E_r : k] < \infty$ for some r .

In characteristic zero, condition (iv) means $[K : k] < \infty$. In positive characteristic it is equivalent to saying that the relative separable closure S of k in K is finite over k and $K^{p^e} \subseteq S$ for some e .

Proof. Assume (iv). The ring $E_r[[\mathbf{X}]]$ is finite over A . For

$$F = \sum_{\alpha} a_{\alpha} \mathbf{X}^{\alpha} \in B,$$

Frobenius gives

$$F^{p^r} = \sum_{\alpha} a_{\alpha}^{p^r} \mathbf{X}^{p^r \alpha} \in E_r[[\mathbf{X}]].$$

Hence F is integral over $E_r[[\mathbf{X}]]$ and therefore over A . In characteristic zero, finite K/k gives module-finiteness directly. An integral extension has both going up and incomparability.

For the converse, argue contrapositively. A transcendental element gives failure of both properties by Proposition 5.33. Suppose therefore that K/k is algebraic but (iv) fails. In characteristic p , for each $r \geq 0$ choose a sequence $c_{r,1}, c_{r,2}, \dots$ such that

$$k(c_{r,1}^{p^r}) \subsetneq k(c_{r,1}^{p^r}, c_{r,2}^{p^r}) \subsetneq \dots$$

In characteristic zero choose one sequence with strictly increasing finite coefficient fields. Interleave the positive-characteristic sequences as $c_1, c_2, \dots$, put $L = k(c_1, c_2, \dots)$, and set

$$g(X) = \sum_{m \geq 1} c_m X^m \in XK[[X]].$$

Then

$$[kL^{p^r} : k] = \infty \quad (r \geq 0).$$

The formal-series transcendence criterion of Cutkosky–Kashcheyeva [9] says exactly that g is transcendental over $k((X))$.

Substitution defines an injection

$$\psi : k[[X, Y]] \longrightarrow K[[X]], \quad F \longmapsto F(X, g(X)).$$

Indeed, a nonzero kernel element, after removal of a common power of X , would be a unit times a monic polynomial in Y by Weierstrass preparation over $k[[X]]$. Evaluation at g preserves the unit and would make g algebraic over $k((X))$.

In $K[[X, Y]]$, the ideal $Q = (Y - g(X))$ is prime and has zero contraction by injectivity. Thus $(0) \subsetneq Q$ disproves incomparability. Going up fails above $(0) \subsetneq (X)$: a prime above (X) and containing Q would contain X , hence $g(X)$ and then Y , although $Y \notin (X)$ in the small ring. Extra variables are carried along coefficientwise. This proves the four-way equivalence.

Finally suppose E_r/k is finite. Any element of the relative separable closure is both separable over k and purely inseparable over its p^r th power, so it lies in E_r ; hence S/k is finite. The finite purely inseparable extension E_r/S has bounded exponent, giving $K^{p^{r+e}} \subseteq S$ for some e . Conversely, $K^{p^e} \subseteq S$ and $[S : k] < \infty$ imply $E_e \subseteq S$ and hence finite E_e/k . $\square$

Theorem 5.35 (Zero-dimensional coefficient classification) [Project result].

Let $R \subseteq T$ be zero-dimensional SFT rings and $n \geq 2$. Put

$$A = R[[X_1, \dots, X_n]], \quad B = T[[X_1, \dots, X_n]].$$

For $q \in \text{Spec } T$, set

$$p = q \cap R, \quad k_q = R/p, \quad K_q = T/q.$$

The following are equivalent:

- (i) $A \subseteq B$ is integral;
- (ii) $A \subseteq B$ has going up;

- (iii) $A \subseteq B$ has incomparability;
- (iv) for every q , the residue-field extension $k_q \subseteq K_q$ satisfies Theorem 5.34; explicitly,

$$[K_q : k_q] < \infty$$

in characteristic zero, while in characteristic $p > 0$ one has

$$[k_q(K_q^{p^{r(q)}}) : k_q] < \infty$$

for some $r(q)$.

Proof. Let $Q \in \text{Spec } B$, set $q = Q \cap T$, $P = Q \cap A$, and $p = P \cap R = q \cap R$. Strong convergence gives

$$q[[\mathbf{X}]] \subseteq Q, \quad p[[\mathbf{X}]] \subseteq P.$$

A chain of primes of A has constant contraction to the zero-dimensional ring R , and a chain in B has constant contraction to T . Quotienting therefore identifies every lifting or comparability problem over q with

$$k_q[[\mathbf{X}]] \subseteq K_q[[\mathbf{X}]].$$

Lifting chains through the quotient maps and contracting them back proves that $A \subseteq B$ has going up, respectively incomparability, exactly when every displayed residue-field map does. By Theorem 5.34, either property is equivalent to (iv).

It remains to prove that (iv) implies integrality. An SFT ring has Noetherian spectrum. A reduced zero-dimensional ring is absolutely flat, so its spectrum is Hausdorff; Noetherian and Hausdorff makes the spectrum finite. If N_R, N_T are the nilradicals, the Chinese remainder theorem gives

$$R/N_R \cong \prod_{p \in \text{Spec } R} R/p, \quad T/N_T \cong \prod_{q \in \text{Spec } T} T/q.$$

The radical identity gives

$$\text{Nil}(A) = N_R[[\mathbf{X}]], \quad \text{Nil}(B) = N_T[[\mathbf{X}]].$$

To justify this without a uniform nilpotence exponent, note that an element of the ordinary extended ideal $N_R A$ is a finite sum with nilpotent coefficients; the finitely generated coefficient ideal is nilpotent. Thus $\sqrt{N_R A} = \sqrt{0}$, while strong convergence identifies the former radical with $N_R[[\mathbf{X}]]$; similarly for T .

Finite products commute with formal power series, so the reduced map is

$$\prod_p (R/p)[[\mathbf{X}]] \longrightarrow \prod_q (T/q)[[\mathbf{X}]],$$

with the q -coordinate induced by $p = q \cap R$. Condition (iv) and Theorem 5.34 make every coordinate extension integral. There are only finitely many coordinates, so the product map is integral: for a fixed element, multiply the finitely many monic coordinate equations after equalizing their degrees.

For $b \in B$, reduced integrality supplies a monic $f(Z) \in A[Z]$ with $f(b) \in \text{Nil}(B)$. Some power of $f(b)$ is zero, so the monic polynomial $f(Z)^m$ annihilates b . Thus B is integral over A . Finally integrality implies going up and incomparability by Cohen–Seidenberg. $\square$

For one formal variable, the zero-dimensional SFT theorem of Condo–Coykendall–Dobbs gives going up and incomparability without the residue restrictions above [6]. The exact classification in Theorem 5.35 is therefore intrinsically a phenomenon of at least two variables.

6. CARDINAL PRIME CHAINS AND THE EXPLICIT FORMAL RINGS

Ordinary Krull dimension and the size of one actual prime chain are different invariants. We use the following notation only in this section.

Definition 6.1 (Notation used in this article). For a ring A , define its *cardinal chain dimension* by

$$\text{cdim } A = \sup\{|\mathcal{C}| : \mathcal{C} \subseteq \text{Spec } A \text{ is totally ordered by inclusion}\}.$$

Thus $\text{dim } A = \infty$ means that finite chains have unbounded length, whereas $\text{cdim } A \geq \aleph_0$ may merely be an unattained supremum. A displayed lower bound $\text{cdim } A \geq \kappa$ will always be accompanied by an actual chain of at least κ primes.

We first give the direct project theorem for $R_p[[X]]$ and $R_0[[X]]$. All coefficient, order-theoretic, compactness, and lifting steps needed for the construction are included.

6.1. The common block construction. Let R denote either R_p or R_0 , let K be its coefficient field, and retain u, c and the filtration $(V_n)_{n \geq 0}$. By Lemma 3.2 and Equations (3.7) and (3.8) in characteristic p , and by Equations (3.20) to (3.22) and (3.25) in characteristic zero, one has

$$V_i V_j \subseteq V_{i+j}, \quad c^n V_n \subseteq R, \quad k[u] \not\subseteq V_E \quad (E \geq 0), \quad (6.1)$$

and every $a \in R$ has a unique expansion

$$a = \sum_{n \geq 0} c^n a_n, \quad a_n \in V_n. \quad (6.2)$$

Moreover $u^r \in V_r$ for every $r \geq 0$. In characteristic p , V_n is Frobenius-stable and $u^{p^a} \in V_1$ for every a ; in characteristic zero, $u^{2^a} \in V_1$ for every a .

Put $q = p$ for $R = R_p$ and $q = 2$ for $R = R_0$, and set $N_L = q^L$.

Lemma 6.2 (Complexity of geometric blocks) [Project result].

Define

$$E_L = \min\{E : u^r \in V_E \text{ for every } 0 \leq r < N_L\}.$$

Then E_L is finite, is nondecreasing, and tends to infinity. Furthermore,

$$E_{L+s} \leq E_L + s(p-1) \quad (R = R_p), \quad (6.3)$$

$$E_{L+s} \leq E_L + s \quad (R = R_0). \quad (6.4)$$

Proof. Finiteness follows from $u^r \in V_r$, and monotonicity is immediate. If (E_L) were bounded, every power of u would lie in one fixed V_E , contrary to Equation (6.1).

In characteristic p , write $0 \leq r < p^{L+s}$ as $r = p^s a + t$, where $0 \leq a < p^L$ and $0 \leq t < p^s$. Frobenius stability gives $(u^a)^{p^s} \in V_{E_L}$. If $t = \sum_{i=0}^{s-1} t_i p^i$, then

$$u^t = \prod_{i=0}^{s-1} (u^{p^i})^{t_i} \in V_{\sum_i t_i} \subseteq V_{s(p-1)}.$$

Multiplicativity proves Equation (6.3). In characteristic zero write $r = a + \sum_{i=L}^{L+s-1} \epsilon_i 2^i$, with $a < 2^L$ and $\epsilon_i \in \{0, 1\}$. Multiplication, using $u^{2^i} \in V_1$, gives $u^r \in V_{E_L+s}$ and proves Equation (6.4). $\square$

Lemma 6.3 (Universal separated scales) [Project result].

There are strictly increasing positive integers L_j, T_j such that, with $H_j = E_{L_j}$ and $N_j = q^{L_j}$,

$$H_j > j^2(H_{j-1} + j), \quad T_j > j^2(T_{j-1} + N_j) \quad (j \geq 2). \quad (6.5)$$

Consequently, for every fixed $a, s \geq 0$,

$$\frac{a + E_{L_{j-1}+s}}{H_j} \longrightarrow 0, \quad \frac{a(T_{j-1} + N_j)}{T_j} \longrightarrow 0. \quad (6.6)$$

Proof. Choose $L_j > L_{j-1}$ recursively, using Lemma 6.2 and $E_L \rightarrow \infty$, and then choose T_j . The two recurrences in Lemma 6.2, followed by Equation (6.5), give Equation (6.6). $\square$

Lemma 6.4 (Support separation and coefficient control) [Project result].

Let $B = R[[X]]$. If $b \in B$ and

$$[X^t]b = \sum_{n \geq 0} c^n b_{t,n},$$

then $b_{t,n} \in V_n$. At any fixed pair of c - and X -degrees, coefficients of a finite product are finite convolutions.

Fix positive integers A and P . For all sufficiently large j the following assertions hold.

(i) A product of A characteristic- p Frobenius blocks

$$X^{PT_s}(1 - (uX)^{PN_s})$$

whose levels are not all j has support disjoint from $[PAT_j, PAT_j + PN_j]$.

(ii) A product of A characteristic-zero blocks

$$X^{T_s}(1 - (uX)^{N_s})$$

whose levels are not all j has support disjoint from $[AT_j, AT_j + N_j]$.

(iii) In either case, a product containing a level greater than j begins beyond the respective target interval. Multiplication by a power series cannot move its support back into the interval.

Proof. The coefficient assertion is Equation (6.2), applied to each X -coefficient. A coefficient of a product at fixed degrees is a finite sum because only finitely many nonnegative degree tuples have the prescribed sum.

A level- s block is supported in $[T_s, T_s + N_s]$, or in its P -multiple in characteristic p . A tuple containing a level greater than j starts at least at T_{j+1} , respectively PT_{j+1} , which is beyond the target by Equation (6.5). If all levels are at most j and one is smaller, its upper endpoint is at most

$$(A-1)(T_j + N_j) + T_{j-1} + N_{j-1} < AT_j$$

for all sufficiently large j . Multiplying this inequality by P proves the Frobenius case. $\square$

Fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$, write ${}^*\mathbb{N} = \mathbb{N}^{\mathbb{N}}/\mathcal{U}$, and put $H = [H_j]_{\mathcal{U}}$. For positive hyperintegers write

$$a \prec b \iff na < b \text{ for every standard } n \geq 1.$$

Let Λ be the interval of Archimedean classes $[d]$ satisfying $1 \prec d \prec H$, ordered in the reverse Archimedean order. For each $\lambda \in \Lambda$, choose a representative $d_\lambda(j)$ with $1 \leq d_\lambda(j) \leq H_j$.

Put $z = uX$, $B = R[[X]]$, $G = B[1/c]$, and $f = 1 - z$. Since $cf = c - cuX \in B$, one has $f \in G$. Define

$$\mathcal{C} = (G/fG)_{\text{red}}, \quad S_L = \sum_{r=0}^{N_L-1} z^r,$$

so that $fS_L = 1 - z^{N_L}$. For $\lambda \in \Lambda$, put

$$Q_\lambda = \sum_{j \geq 1} c^{d_\lambda(j)} X^{T_j} S_{L_j} \in K[[c]][[X]], \quad (6.7)$$

$$g_\lambda = fQ_\lambda = \sum_{j \geq 1} c^{d_\lambda(j)} X^{T_j} (1 - z^{N_j}) \in B, \quad (6.8)$$

and let x_λ be the image of g_λ in $\mathcal{C}$. Indeed $u^{N_j} \in V_1 \subseteq V_{d_\lambda(j)}$, so both coefficients of each block in Equation (6.8) belong to R . Since $T_j \rightarrow \infty$, every X -coefficient receives contributions from only finitely many blocks. Thus both displayed series are well defined. The ambient series Q_λ need not belong to G ; only $g_\lambda \in B$ is asserted.

Lemma 6.5 (Archimedean depth estimates) [Project result].

Let $\lambda_1 < \dots < \lambda_\ell$ in Λ , put $d_i(j) = d_{\lambda_i}(j)$, and fix $m_i \geq 1$. With

$$\Delta_j = \sum_{h=1}^{\ell} m_h d_h(j), \quad \delta_i(j) = d_i(j) - \sum_{h>i} m_h d_h(j), \quad (6.9)$$

one has

$$\Delta_j = o_{\mathcal{U}}(H_j), \quad \delta_i(j) \rightarrow_{\mathcal{U}} \infty.$$

Proof. Every $d_i \prec H$, so every fixed finite linear combination of the d_i is still $\prec H$. Since the order on Λ is reversed, $d_h \prec d_i$ for $h > i$. Given a standard C , on a $\mathcal{U}$ -large set d_i dominates $C + \sum_{h>i} m_h d_h$, proving the second assertion. $\square$

Recall that a finite sequence $y_1, \dots, y_\ell$ in a ring is *pseudo-regular* if there are no $m_i \geq 0$ and a_i in the ring for which

$$\prod_{h=1}^{\ell} y_h^{m_h} + \sum_{i=1}^{\ell} a_i y_i^{m_i+1} \prod_{h<i} y_h^{m_h} = 0.$$

Proposition 6.6 (Finite pseudo-regularity) [Project result].

Given elements of Λ satisfying

$$\lambda_1 < \dots < \lambda_\ell,$$

the elements $x_{\lambda_1}, \dots, x_{\lambda_\ell}$ form a pseudo-regular sequence in $\mathcal{C}$.

Proof. Write $x_i = x_{\lambda_i}$ and $g_i = g_{\lambda_i}$. Suppose the sequence is singular. By the definition of pseudo-regularity, there are $m_i \in \mathbb{N}$ and $a_i \in \mathcal{C}$ such that

$$\prod_{h=1}^{\ell} x_h^{m_h} + \sum_{i=1}^{\ell} a_i x_i^{m_i+1} \prod_{h<i} x_h^{m_h} = 0. \quad (6.10)$$

We may assume $m_i \geq 1$. If $m_k = 0$, multiply the relation by x_k ; for $i < k$ absorb the extra x_k into a_i , and repeat.

Lift a_i to $y_i \in G$ and let $\Phi \in G$ be the corresponding left side in the g_i . Its class in G/fG is nilpotent. Hence $\Phi^P = fZ$ for some $Z \in G$, where $P = p^k$ in characteristic p and P is an arbitrary sufficiently large positive integer in characteristic zero. Choose $b, D \geq 0$ such that

$$Y_i = c^b y_i \in B, \quad c^D Z \in B, \quad D \geq Pb. \quad (6.11)$$

Put $M = \sum_h m_h$ and use $\Delta_j, \delta_i(j)$ from Equation (6.9).

Characteristic p . Freshman's dream gives

$$c^D Z = \frac{c^D}{f} \prod_h g_h^{Pm_h} + \sum_i \frac{c^{D-Pb} Y_i^P}{f} g_i^{P(m_i+1)} \prod_{h<i} g_h^{Pm_h}. \quad (6.12)$$

Moreover

$$g_h^P = \sum_s c^{Pd_h(s)} X^{PT_s} (1 - z^{PN_s}). \quad (6.13)$$

Fix j in a $\mathcal{U}$ -large set on which all the inequalities below hold, and inspect

$$[c^{D+P\Delta_j} X^{PMT_j+r}] c^D Z, \quad 0 \leq r < PN_j. \quad (6.14)$$

By Lemma 6.4, the only leading selection meeting this window is the all-level- j selection. Its polynomial factor is

$$\frac{(1 - z^{PN_j})^M}{f} = f^{PMN_j-1} = S_{L_j+k}(1 - z^{PN_j})^{M-1},$$

so its contribution to Equation (6.14) is u^r .

Consider a selection in the i th correction term. A level greater than j is excluded by support. The all-level- j selection has c -depth

$$D + P\Delta_j + P(\delta_i(j) - b) > D + P\Delta_j$$

for $\mathcal{U}$ -almost all j , and hence cannot contribute at the target layer. It remains to consider a selection containing a lower level. Let a be the number of level- j blocks, let C_0 be the sum, with multiplicity, of the N_s over lower blocks, and choose an integer ρ depending only on Equation (6.10) such that $C_0 \leq \rho N_{j-1}$ and $a \leq \rho$. After division by f , its block factor is

$$(1 - z^{PN_j})^a f^{PC_0-1}. \quad (6.15)$$

Choose e with $p^e \geq \rho$. Every coefficient of Equation (6.15) lies in

$$V_{\rho+E_{L_{j-1}+k+e}}. \quad (6.16)$$

Indeed, the first factor costs at most ρ copies of $u^{PN_j} \in V_1$, while every power of u occurring in the second has exponent less than $P\rho N_{j-1} \leq p^{L_{j-1}+k+e}$.

If Γ is the selected block depth before multiplication by P , a contribution at Equation (6.14) requires $q = \Delta_j + b - \Gamma \geq 0$. By Equation (6.2) and Frobenius stability,

$$[c^{Pq} X^t] Y_i^P \in V_q.$$

Thus the sum $\beta_{j,r}$ of all surviving correction contributions lies in

$$V_{Q_j}, \quad Q_j = \Delta_j + b + \rho + E_{L_{j-1}+k+e}. \quad (6.17)$$

There are only finitely many selections and convolutions at the fixed degrees in Equation (6.14). Since $c^D Z \in B$, Equation (6.2) yields

$$u^r + \beta_{j,r} \in V_{D+P\Delta_j}.$$

It follows, for every $0 \leq r < PN_j$, that

$$u^r \in V_{\max\{D+P\Delta_j, Q_j\}},$$

and hence

$$E_{L_j+k} \leq \max\{D + P\Delta_j, Q_j\}. \quad (6.18)$$

By Equations (6.3) and (6.6) and Lemma 6.5, the right side is less than $H_j = E_{L_j}$ on a $\mathcal{U}$ -large set. This contradicts $E_{L_j} \leq E_{L_j+k}$.

Characteristic zero. Put

$$A_0 = \prod_h g_h^{m_h}, \quad B_i = g_i^{m_i+1} \prod_{h<i} g_h^{m_h}.$$

The full multinomial expansion is

$$\Phi^P = \sum_{e_0+\dots+e_\ell=P} \binom{P}{e_0, \dots, e_\ell} A_0^{e_0} \prod_{i=1}^\ell (y_i B_i)^{e_i}. \quad (6.19)$$

Let $E = \sum_{i \geq 1} e_i$. In an all-level- j selection its block depth is

$$P\Delta_j + \sum_{i=1}^\ell e_i \delta_i(j). \quad (6.20)$$

After replacing y_i by $c^{-b} Y_i$, every term with $E > 0$ begins above the target layer $D + P\Delta_j$, because

$$\sum_i e_i \delta_i(j) - bE > 0$$

for $\mathcal{U}$ -almost all j . Hence the only all-level- j term that can contribute is $e_0 = P$. Inspect

$$[c^{D+P\Delta_j} X^{PMT_j+r}] c^D Z, \quad 0 \leq r < N_j. \quad (6.21)$$

The pure all-level- j term contributes u^r , since

$$\frac{(1 - z^{N_j})^{PM}}{f} = S_{L_j}(1 - z^{N_j})^{PM-1}. \quad (6.22)$$

Pure leading selections at other levels and all selections containing a level greater than j are excluded by Lemma 6.4.

Every remaining selection contains a lower block. If a blocks have level j and the lower levels are $s_1, \dots, s_t < j$, its block factor after division by f is

$$(1 - z^{N_j})^a f^{t-1} \prod_{\nu=1}^t S_{L_{s_\nu}}. \quad (6.23)$$

Choose a relation-dependent ρ bounding the total number of blocks in Equation (6.19), and then e with $2^e > 2\rho$. The lower factor in Equation (6.23) involves only powers u^v with

$$v < \rho(N_{j-1} + 1) < 2^{L_{j-1}+e},$$

while the first factor costs at most ρ copies of $u^{N_j} \in V_1$. Hence every coefficient of Equation (6.23) lies in

$$V_{\rho+E_{L_{j-1}+e}}. \quad (6.24)$$

If Γ is the selected block depth, contribution at the target layer requires

$$q = P\Delta_j + bE - \Gamma \geq 0, \quad q \leq P\Delta_j + Pb.$$

By finite convolution and Equation (6.2), every c^q -coefficient of $\prod_i Y_i^{e_i}$ lies in V_q . Thus all correction contributions sum to an element of

$$V_{Q'_j}, \quad Q'_j = P\Delta_j + Pb + \rho + E_{L_{j-1}+e}. \quad (6.25)$$

Equations (6.21), (6.22) and (6.25) and Equation (6.2) imply

$$E_{L_j} \leq \max\{D + P\Delta_j, Q'_j\} < H_j = E_{L_j}$$

on a $\mathcal{U}$ -large set, by Equations (6.4) and (6.6) and Lemma 6.5. This contradiction proves pseudo-regularity in characteristic zero and completes the proof. $\square$

6.2. From finite patterns to one huge chain.

Lemma 6.7 (The Archimedean interval is an η_1 -order) [Project result].

Modulo Archimedean equivalence

$$a \asymp b \iff a \leq nb \text{ and } b \leq ma \text{ for some standard } m, n,$$

the interval $1 \prec d \prec H$ in ${}^\mathbb{N}$ is an η_1 -order. Every class in it has a representative satisfying $1 \leq d(j) \leq H_j$ for every j .*

Proof. A nonprincipal ultrapower over $\mathbb{N}$ is countably saturated. One may see this directly from countable incompleteness of $\mathcal{U}$: for a countable finitely satisfiable type, choose a witness to its first n conditions and use a decreasing sequence of members of $\mathcal{U}$ with empty intersection to make the witness index tend to infinity modulo $\mathcal{U}$.

Let $([a_i])_{i \geq 1}$ and $([b_i])_{i \geq 1}$ be countable families in the interval with $[a_i] < [b_j]$ for all i, j . The type

$$na_i < x, \quad nx < b_i \quad (i, n \geq 1), \quad (6.26)$$

together with $1 \prec x \prec H$, is countable and finitely satisfiable. Indeed, for a finite subsystem take the largest lower Archimedean class and the smallest upper class. The latter dominates every standard multiple of the former, so a sufficiently large standard multiple of a representative of the lower class realizes all finitely many inequalities. Saturation realizes Equation (6.26); its class interpolates the two families. Clipping a representative to $[1, H_j]$ changes it only off a member of $\mathcal{U}$ and hence does not change its class. $\square$

Lemma 6.8 (Cuts in an η_1 -order) [Project result].

Every η_1 -order has at least $2^{\aleph_1}$ distinct nonempty initial segments.

Proof. Construct intervals (a_s, b_s) for $s \in 2^{<\omega_1}$. At a successor stage choose two disjoint subintervals

$$a_s < a_{s0} < b_{s0} < a_{s1} < b_{s1} < b_s.$$

At a limit stage $\alpha < \omega_1$, the branch below a node has only countably many earlier endpoints. The η_1 property supplies two points strictly between all earlier left and right endpoints, and hence a new nonempty interval. For $\tau \in 2^{\omega_1}$ put

$$D_\tau = \bigcup_{\alpha < \omega_1} (-\infty, a_{\tau|_\alpha}).$$

These are initial segments. If two branches first split at a node s , all later intervals on the 0-branch lie to the left of a_{s1} , whereas a later left endpoint on the 1-branch lies to its right. Their cuts are therefore distinct. Discarding an empty cut, if one occurs, does not change the cardinality. $\square$

We use Lombardi's theorem [17, Theorem 5]: a finite sequence $(x_1, \dots, x_r)$ is pseudo-regular precisely when there are primes

$$Q_0 \subsetneq Q_1 \subsetneq \dots \subsetneq Q_r, \quad x_i \in Q_i \setminus Q_{i-1}.$$

The next lemma makes the compatibility step, which is not part of that finite theorem, explicit.

Lemma 6.9 (Patch-compactness globalization) [Project result].

Let A be a ring, let Λ be a linear order, and let $(x_\lambda)_{\lambda \in \Lambda}$ be a family such that every finite increasing subfamily is pseudo-regular. Then there are primes $(P_\lambda)_{\lambda \in \Lambda}$ satisfying, whenever $\lambda < \mu$,

$$P_\lambda \subsetneq P_\mu, \quad x_\lambda \in P_\lambda, \quad x_\mu \notin P_\lambda. \quad (6.27)$$

Proof. Give $\text{Spec } A$ its compact Hausdorff patch topology [19, Tag 0901]; Tychonoff's theorem makes $(\text{Spec } A)^\Lambda$ compact. Membership $x \in P_\lambda$ and nonmembership are clopen. The relation $P_\lambda \subseteq P_\mu$ is closed because its complement is

$$\bigcup_{a \in A} \{(P_\nu) : a \in P_\lambda, a \notin P_\mu\},$$

a union of clopen rectangles.

Impose the closed conditions

$$x_\lambda \in P_\lambda, \quad P_\lambda \subseteq P_\mu, \quad x_\mu \notin P_\lambda \quad (\lambda < \mu).$$

They have the finite-intersection property. For finitely many indices $\lambda_1 < \dots < \lambda_r$, Lombardi's theorem supplies

$$Q_0 \subsetneq Q_1 \subsetneq \dots \subsetneq Q_r, \quad x_{\lambda_i} \in Q_i \setminus Q_{i-1}.$$

Assign $P_{\lambda_i} = Q_i$. If $i < j$, then $Q_i \subseteq Q_{j-1}$ and $x_{\lambda_j} \notin Q_{j-1}$, so every required marker condition holds. Compactness gives a simultaneous choice. Finally, $x_\mu \in P_\mu \setminus P_\lambda$ forces every displayed inclusion to be strict. $\square$

Lemma 6.10 (From the indexed chain to $2^{\aleph_1}$ primes) [Project result].

Under the hypotheses of Lemma 6.9, if Λ is an η_1 -order, then $\text{Spec } A$ contains a chain of cardinality at least $2^{\aleph_1}$.

Proof. For each nonempty initial segment $D \subseteq \Lambda$, put

$$Q_D = \bigcup_{\lambda \in D} P_\lambda.$$

A union of a chain of prime ideals is prime, and the Q_D form a chain. If $D \subsetneq E$ and $\mu \in E \setminus D$, then every $\lambda \in D$ satisfies $\lambda < \mu$. Hence

$$x_\mu \in Q_E \setminus Q_D$$

by Equation (6.27). Distinct cuts therefore give distinct primes, and Lemma 6.8 applies. $\square$

Lemma 6.11 (Lifting the chain) [Project result].

Every strict chain in $\text{Spec } \mathcal{C}$ lifts, with the same order type, to a strict chain in $\text{Spec } B$.

Proof. Prime ideals of a ring and of its reduction correspond by inverse image, so a chain in $\mathcal{C}$ lifts strictly to G/fG , and then to primes of G containing f . Localization gives an inclusion-preserving bijection between $\text{Spec } G$ and the primes of B disjoint from $\{1, c, c^2, \dots\}$. Contraction preserves strictness because extension back to G recovers the original prime. $\square$

Theorem 6.12 (Huge one-variable prime chains) [Project result].

For every prime p , the explicit direct rings satisfy

$$\text{cdim } R_p[[X]] \geq 2^{\aleph_1}, \quad \text{cdim } R_0[[X]] \geq 2^{\aleph_1}. \quad (6.28)$$

In particular both one-variable formal power-series rings have infinite ordinary Krull dimension. Under the continuum hypothesis,

$$\text{cdim } R_p[[X]] = \text{cdim } R_0[[X]] = 2^{\aleph_1}. \quad (6.29)$$

Proof. Lemma 6.7 supplies the η_1 -indexed family used in Equation (6.8). By Proposition 6.6, every finite increasing subfamily of its image in $\mathcal{C}$ is pseudo-regular. Lemmas 6.9 to 6.11 give Equation (6.28).

Both R_p and R_0 are countable. Indeed their coefficient fields and filtration pieces are countable, the graded rings are countable unions of countable sets, and localization preserves countability. Therefore

$$|R_p[[X]]| = |R_0[[X]]| = \aleph_0^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c},$$

and each spectrum has cardinality at most $2^{\mathfrak{c}}$. Under CH, $\mathfrak{c} = \aleph_1$, so this upper bound and Equation (6.28) give Equation (6.29). $\square$

Remark 6.13 (Status of the direct theorem). Theorem 6.12 is a project theorem proved here, not one quoted from the literature. The proof above includes the lower-block filtration estimates and the compactness argument needed for the transfinite passage. It has not received external peer review; that distinction is retained explicitly.

6.3. Published explosion and immediate consequences. The earlier literature first produced uncountable chains over non-SFT coefficients [12]; the sharp general result used here is the following.

Theorem 6.14 (Toan–Kang [20, 21]). *If a commutative ring D is not SFT, then $D[[Y]]$ contains a chain of prime ideals of cardinality at least $2^{\aleph_1}$. If a non-SFT maximal ideal M is the radical of a countably generated ideal, the chain may be chosen between*

$$MD[[Y]] \quad \text{and} \quad M[[Y]].$$

The same local conclusion holds for a non-SFT maximal ideal in a zero-dimensional quasilocal ring or a one-dimensional quasilocal domain. Under CH the universal lower bound $2^{\aleph_1}$ is optimal.

The 2018 paper proves the domain version; the 2020 paper proves the stated arbitrary-ring theorem. We quote this published result and prove each application below from the non-SFT conclusions already established.

Corollary 6.15 [Project result].

Let S be R_p or R_0 . Then:

- (i) $S[[X]]$ has infinite ordinary dimension and contains a prime chain of cardinality at least $2^{\aleph_1}$;
- (ii) $S[[X, Y]]$ contains such a chain;
- (iii) $(S[X])[[Y]]$ contains such a chain.

For either ring, every nonempty finite-variable formal ring has cardinality $\mathfrak{c}$; under CH its spectrum and its cardinal chain dimension therefore both have cardinality $2^{\mathfrak{c}}$.

Proof. Part (i) is Theorem 6.12.

By Theorem 4.14, each $S[[X]]$ is non-SFT. Apply theorem 6.14 and use the canonical identification $S[[X]][[Y]] \cong S[[X, Y]]$ to prove (ii). Likewise $S[X]$ is non-SFT by Theorem 3.1; applying theorem 6.14 proves (iii).

The rings R_p and R_0 are countable, and the monomials in any fixed positive finite number of formal variables form a countable set. A formal series is therefore a function from a countable set to a countable set, giving the upper bound $\mathfrak{c}$. Binary coefficient sequences give the reverse bound. Under CH, the chain lower bound is $2^{\aleph_1} = 2^{\mathfrak{c}}$, while every spectrum is a subset of the power set of a ring of size $\mathfrak{c}$. Equality follows for both the spectrum cardinality and cdim . $\square$

6.4. A countable non-SFT domain.

Proposition 6.16 (Countable valuation example) [Project result].

Let

$$K = \bigcup_{n \geq 1} \mathbb{F}_2(t^{1/n!}),$$

let v be the t -adic valuation, and put

$$V = \{0\} \cup \{x \in K^\times : v(x) \geq 0\}, \quad \mathfrak{m} = \{x \in V : v(x) > 0\}.$$

Then V is a countable one-dimensional local valuation domain and is not SFT. Its ring $V[[X]]$ has cardinality $\mathfrak{c}$ and contains an actual prime chain of cardinality at least $2^{\aleph_1}$, which may be chosen between $\mathfrak{m}V[[X]]$ and $\mathfrak{m}[[X]]$.

Proof. Every field $\mathbb{F}_2(t^{1/n!})$ is countable, so K and V are countable. The value group is

$$\bigcup_{n \geq 1} \frac{1}{n!} \mathbb{Z} = \mathbb{Q}.$$

It has rank one, so the only primes of V are (0) and $\mathfrak{m}$.

Suppose $\mathfrak{m}$ had SFT data with a finite ideal $J \subseteq \mathfrak{m}$ and exponent N . A finitely generated ideal in a valuation domain is principal; write $J = (b)$ and $v(b) = \alpha > 0$. The rational power

$$x = t^{\alpha/(2N)}$$

belongs to K and lies in $\mathfrak{m}$, but $v(x^N) = \alpha/2 < \alpha$. Hence $x^N \notin (b)$, a contradiction. Thus V is non-SFT.

The usual sequence argument gives $|V[[X]]| = \mathfrak{c}$. Since V is countable, $\mathfrak{m}$ is countably generated as an ideal. The local form of theorem 6.14 gives the asserted vertical chain. $\square$

6.5. Two exact cardinal-dimension realizations. The next examples clarify why an infinite ordinary dimension does not by itself determine the cardinal size of a chain.

Proposition 6.17 (An unattained $\aleph_0$) [Project result].

There is a countable Noetherian (hence VSFT) ring R such that

$$\text{cdim } R[[X]] = \aleph_0,$$

and $R[[X]]$ is again Noetherian and VSFT, but it has no countably infinite chain of prime ideals.

Proof. Let $k = \mathbb{Q}$, put

$$A = k[x_1, x_2, \dots], \quad \mathfrak{p}_i = (x_{2i-1}, \dots, x_{2i-1}),$$

and let

$$S = \bigcap_{i \geq 1} (A \setminus \mathfrak{p}_i), \quad R = S^{-1}A.$$

This is Nagata's standard countable Noetherian ring of infinite dimension [19, Tag 02JC]. Hence R has finite prime chains of arbitrarily large length. The formal ring $B = R[[X]]$ is Noetherian, and the quotient $B/(X) \cong R$ lifts all those finite chains. Every ideal in either Noetherian ring is finitely generated, so it is VSFT with exponent one and with the ideal itself as witness.

Every chain in a Noetherian ring is finite. Indeed, its union is a prime P ; localization embeds the chain in $\text{Spec } B_P$. The maximal ideal of the Noetherian local ring B_P is finitely generated, so the Krull height theorem gives $\dim B_P < \infty$. An infinite chain would contain finite subchains of unbounded length, a contradiction. Thus the finite chain cardinalities in B are unbounded but no infinite cardinality is attained, and

$$\text{cdim } B = \sup_{n < \omega} n = \aleph_0.$$

Replacing $\mathbb{Q}$ by $\mathbb{F}_p$ gives the same example in positive characteristic. $\square$

Proposition 6.18 (Exact $\aleph_1$ under CH) [Project result].

Assume CH. In positive characteristic and in equicharacteristic zero there is a VSFT valuation domain R such that $R[[X]]$ is SFT and

$$\text{cdim } R[[X]] = \aleph_1.$$

The coefficient domain in this construction is itself infinite-dimensional.

Proof. Let $k = \mathbb{F}_p$ or $\mathbb{Q}$ and give

$$\Gamma = \bigoplus_{\alpha < \omega_1} \mathbb{Z}e_\alpha$$

the reverse lexicographic order: the coefficient at the largest nonzero index determines the sign. Put

$$K = \text{Frac}(k[\Gamma]), \quad R = \{z \in K : v(z) \geq 0\},$$

where v is the monomial valuation. The convex subgroups are precisely

$$\Gamma_\beta = \bigoplus_{\alpha < \beta} \mathbb{Z}e_\alpha \quad (\beta \leq \omega_1).$$

The prime-convex-subgroup correspondence for valuation domains therefore makes $\text{Spec } R$ a chain of cardinality $\aleph_1$.

We verify the stronger VSFT property directly. Let I be a nonzero proper ideal and let $S_I = \{v(a) : 0 \neq a \in I\}$. Among the largest support indices of values in S_I , choose the least one, say β ; among the positive e_β -coefficients occurring at that layer, choose the least, say n . Choose $d \in I$ at that layer with coefficient n . If $a, b \in I$ are nonzero, then either one of their values has a larger leading index, or both leading e_β -coefficients are at least n . In either case

$$v(ab) \geq v(d).$$

Thus $I^2 \subseteq dR \subseteq I$. The zero and unit ideals are immediate, so R is VSFT, hence SFT, and it is Prüfer. Kang and Park's theorem [15] makes $R[[X]]$ SFT.

The quotient $R[[X]]/(X) \cong R$ lifts the $\aleph_1$ -chain. Also $|\Gamma| = |R| = \aleph_1$, and CH gives

$$|R[[X]]| = (\aleph_1)^{\aleph_0} = \aleph_1.$$

In an SFT ring every prime P is the radical of a finitely generated ideal: an SFT witness $J \subseteq P$ gives $\sqrt{J} = P$. A ring of cardinality $\aleph_1$ has only $\aleph_1$ finite subsets, hence the SFT ring $R[[X]]$ has at most $\aleph_1$ primes. The lower and upper bounds prove the claimed equality. $\square$

Remark 6.19. The construction in Proposition 6.18 does not answer the stronger question whether a *finite-dimensional* coefficient ring can satisfy $\text{cdim } R[[X]] = \aleph_1$ under CH. A non-SFT coefficient ring cannot do so, because theorem 6.14 forces the strictly larger lower bound $2^{\aleph_1}$. Any such finite-dimensional example would have to be SFT; the project leaves that narrower question open.

7. EXACT BOUNDARIES AND NONCLAIMS

This final section prevents the proved implications from being read more broadly than their hypotheses allow. Each item is a logical boundary of the preceding proofs, not a conclusion drawn from numerical evidence.

Table 6: Questions not settled by the results in this article.

Question	Exact proved boundary
Classify all SFT rings R for which $R[X]$ is SFT.	Theorem 2.15 is an exact reduction to uppers to zero over all residue domains; it is not an intrinsic structural classification.
Does normal SFT imply polynomial stability in arbitrary dimension?	The answer is positive in dimension at most one, for normal going-down domains [18][4, Theorem 4.1], and under the finite-dimensional residual N-1 hypotheses proved in the polynomial section. No broader assertion is made.
Does $R[[X]]$ SFT imply $R[[X, Y]]$ SFT?	The problem reduces to domains and to horizontal primes; the trace and moving-axis theorems isolate several positive strata. They do not prove the universal implication.
Does $D[[Y]]$ having the SFT, t -SFT, or VSFT property force $D[X]$ to have the same property?	Both polynomial and formal extension descend to D for all three properties by Corollary 4.7, but ascent from D to $D[X]$ is not automatic. The t -SFT implication is positive when D is integrally closed by theorem 4.3 [14, Theorem 2.4], and all three are positive when D is Noetherian. No general formal-to-polynomial implication is proved here.
Can a finite-dimensional normal SFT domain have an infinite-dimensional one-variable formal-series ring?	Generic-fibre uniformity is equivalent to finite dimension, and the valuation-conductor and one-dimensional reductions give necessary conditions. The constructed LPVD is a stress test, not a counterexample.
Is the direct bound $2^{\aleph_1}$ the exact cardinal chain dimension in ZFC?	Theorem 6.12 proves the lower bound and proves equality under CH. In ZFC the general upper bound is 2^c ; no equality between these cardinals is assumed.

7.1. Safe implication chart. For rings, the unconditional implications among the three finite-type conditions used here are

$$\text{VSFT} \implies \text{SFT},$$

and, within the category of domains,

$$\text{VSFT} \implies \text{SFT} \implies t\text{-SFT}.$$

The first converse fails already for R_p , while the second converse fails for $k[Z_1, Z_2, \dots]$ by Proposition 4.5. For domains containing $\mathbf{Q}$, SFT and VSFT agree by Coykendall–Dutta’s polarization theorem [8, Theorem 5.1].

For extension types, the proved global picture is

$$\begin{aligned} A \subseteq B \text{ module-finite} &\implies A \text{ SFT} \iff B \text{ SFT}, \\ A \rightarrow B \text{ faithfully flat} &\implies B \text{ SFT} \implies A \text{ SFT}, \\ A \subseteq B \text{ bounded radicial} &\implies A \text{ SFT} \implies B \text{ SFT}, \\ A \twoheadrightarrow B &\implies A \text{ SFT} \implies B \text{ SFT}. \end{aligned}$$

Polynomial extension is flat, so ordinary SFT gives descent but not ascent: the explicit rings prove that the reverse implication fails even for one-dimensional local domains. Formal extension gives ordinary SFT and VSFT descent by quotienting at the formal variables [8, Theorem 4.6]. For t -SFT, neither descent is a quotient argument; the exact contraction proof is Proposition 4.6. Formal SFT ascent is governed by the horizontal-prime criterion. The explicit coefficient rings now give negative zero-to-one examples by Theorem 4.14; the universal one-to-two formal ascent problem is still kept separate.

7.2. What the polynomial examples do and do not transfer. Let S be R_p or R_0 . The quotient maps

$$S[[Y_1, \dots, Y_s]][X] \twoheadrightarrow S[X]$$

prove that the source is non-SFT. By contrast, substitution $X \mapsto u$ in the polynomial obstruction is a finite sum. There is generally no ring homomorphism

$$S[[X]] \longrightarrow \text{Frac}(S), \quad \sum_{i \geq 0} a_i X^i \longmapsto \sum_{i \geq 0} a_i u^i,$$

because the infinite sum has no prescribed meaning in $\text{Frac}(S)$. Thus the non-SFT upper to zero in $S[X]$ cannot be transported to $S[[X]]$ by formal evaluation. The later direct proof instead uses the polynomial blocks $X^N - u^N$ inside the formal ring itself. Thus the warning about evaluation remains correct even though the formal entries are now settled.

7.3. Proof-only character of the compendium. The dimension arguments use prime chains, localization, integral-extension theorems, and generic formal fibres. The examples use filtrations, coefficient comparison, Frobenius or norm identities, and pullbacks. None of the conclusions in the summary tables depends on a computer enumeration, an automated bounded search, random sampling, or experimental extrapolation.

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